

Technical Specification 43-50
Issue 1: 1984

**132kV Single Circuit Overhead
Lines on Wood Poles**

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AMENDMENT 1, 1993

Technical Specification 43-50 Issue 1, 1984

132kV single circuit overhead lines on wood poles

Page 7, FOREWORD

Insert new paragraph at end of this section:

"This Technical Specification has been amended to introduce the changes necessary to ensure that it meets the requirements of Clause 11 of the Utilities Supply and Works Contracts Regulations 1992 (S.I. 1992 No. 3279)"

Page 7, SCOPE Add the following text:

"Since this Specification is based on, "The Electricity (overhead lines) Regulations 1970, S.I. 1970 No 1355" which have been replaced by: "The Electricity Supply Regulations 1988, S.I. 1988 No 1057" and amendments, the users of this Specification should ensure that overhead lines designed to the requirements of this Specification comply with the later Regulations."

Page 8, REFERENCES

Delete the complete clause and **insert** the following:

- | | |
|-------------|---|
| BS 215 | Aluminium conductors and aluminium conductors steel-reinforced for overhead power transmission:
Part 2, Aluminium conductors steel reinforced
(Associated Standard, IEC 1089 round wire concentric lay overhead electrical stranded conductors) |
| BS 1990 | Wood poles for overhead power and telecommunication lines |
| EA TS 43-8 | Overhead line clearances |
| EA TS 43-88 | Treatment of wood poles and associated timber for overhead lines. |
| EA TS 43-90 | Anti-climbing devices and safety signs for H.V. lines up to and including 400 kV |

- EA TS 43-91 Stay strands and stay fittings for overhead lines
- EA TS 43-92 Conductor terminations, joints and insulator binds for overhead lines up to and including 132 kV
- EA TS 43-93 Line insulators
- EA TS 43-95 Steelwork for overhead lines
- EA TS 43-96 Fasteners and washers for wood pole overhead lines
- ER L 38 Protective greasing of PVC-covered copper and bare and PVC-covered aluminium-based overhead line conductors.
(Associated Standard, IEC 1089)
- S.I. 1355 The Electricity (Overhead lines) Regulations, 1970
- S.I. 1988 The Electricity Supply Regulations 1988
No 1057 (and amendments)

General

Where the term "ESI Standard" is used **replace** with "EA Technical Specification"

Page 9, Clause 4 Design data

Add the following text:

"The design requirements should where necessary be modified to comply with the requirements of The Electricity Supply Regulations 1988, S.I. 1988 No 1057 and The Electricity Supply (Amendment) Regulations 1990, S.I. 1990 No 390.

Page 10, Clause 5.2

Replace "When fully seasoned and fabricated, poles shall be creosoted in accordance with BS 913 Rueping process. An alternative method of preservative may be applied in accordance with BS 3452 and BS 4072 if specified by the purchaser"

with "...shall be creosoted in accordance with EA TS 43-88. Alternatively Copper/Chrome/Arsenic preservation may be applied in accordance with the same specification if specified by the purchaser".

Page 11, Clause 5.9 danger plates

Delete existing title and text and **replace** with the following:

"Safety signs shall comply with the Electricity Supply Regulations 1988, S.I. 1988 No 1057, The Electricity Supply (Amendment) Regulations 1990, S.I. 1990 No 390, and be shall fitted be in accordance with EA TS 43-90"

Page 14, Clause 6.13 Minimum height of line conductors and clearances

Delete reference to Electricity Regulations and **replace** by "Electricity Supply Regulations 1988"

Page 14, Clause 6.14 Pole mounted cable termination

Delete existing text:

"in accordance with ESI Standard 43-89 or other appropriate Standard".

Insert a full stop after:

"connected to earth".

INDEX

	<i>Page</i>
Foreword	7
1. Scope	7
2. Definitions	7
3. References	8
4. Design Data	9
4.1 Conductors	9
4.2 Crossarm Assemblies	9
4.3 Supports	10
4.4 Insulators and Fittings	10
4.5 Foundations	10
5. Materials	10
5.1 Conductors	10
5.2 Wood Poles	10
5.3 Crossarm Steelwork	11
5.4 Insulators	11
5.5 Insulator Fittings	11
5.6 Conductor Fittings	11
5.7 Stays	11
5.8 Fasteners and Washers	11
5.9 Danger Plates	11
6. Technical Requirements for the Erection of Lines	11
6.1 Span Lengths	11
6.2 Supports	11
6.3 Maximum Angles of Deviation	11
6.4 Wind Loading Spans and Foundations	11
6.5 Capacities of Intermediate Crossarm Assemblies	12
6.6 Vertical Conductor Loads and Weight Span	12
6.7 Staying Arrangements and Strut Loading for Angle and Terminal Supports	13
6.8 Conductor Design and Erection Sags and Tensions	13
6.9 Pilot Insulators	13
6.10 Bonding	13
6.11 Provision of Danger Plates	14
6.12 Precautions against Access	14
6.13 Minimum Height of Line Conductors and Clearances	14
6.14 Pole Mounted Cable Termination	14
6.15 Line Termination on Steel Tower	14
6.16 Downleads	14
6.17 Downlead and Jumper Clearances	14

Index (Cont'd)

	<i>Page</i>
TABLES	
1. Windspan/Pole/Foundation Selection Chart — Single Supports . . .	15
2. Windspan/Pole/Foundation Selection Chart — 'H' Supports	18
3. Ordinates for Plotting Uplift and Ground Clearance Curves — Basic Span 130 m	19
4. Design Sag/Tension Table for 175 mm ² (30 + 7/2.79) ACSR 'Lynx' — Basic Span 130 m	20
5. Design Sag/Tension Table for 175 mm ² (18 + 1/3.61) and 200 mm ² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' — Basic Span 130 m	20
6. Pretension and Erection Table for 175 mm ² (30 + 7/2.79) ACSR 'Lynx' — Basic Span 130 m	21
7. Pretension and Erection Table for 175 mm ² (18 + 1/3.61) and, 200 mm ² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' — Basic Span 130 m	22
8. Ordinates for Plotting Uplift and Ground Clearance Curves — Basic Span 170 m	23
9. Design Sag/Tension Table for 175 mm ² (30 + 7/2.79) ACSR 'Lynx' — Basic Span 170 m	24
10. Design Sag/Tension Table for 175 mm ² (18 + 1/3.61) and 200 mm ² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' — Basic Span 170 m	24
11. Pretension and Erection Table for 175 mm ² (30 + 7/2.79) ACSR 'Lynx' — Basic Span 170 m	25
12. Pretension and Erection Table for 175 mm ² (18 + 1/3.61) and 200 mm ² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' — Basic Span 170 m	26
NOMOGRAM	
1. Mid-Span Spacing/Maximum Span Relationship Adjacent Angles	27
CHARTS	
1. Stay Angles	28
2. Stayed Pole Design	29
3. Stay Sizes and Strut Loads for 175 mm ² (30 + 7/2.79) ACSR 'Lynx'	30
4. Stay Sizes and Strut Loads for 175 mm ² (18 + 1/3.61) ACSR 'Caracal'	31
5. Stay Sizes and Strut Loads for 200 mm ² (18 + 1/3.86) ACSR 'Jaguar'	32

Index (Cont'd)

	<i>Page</i>
ILLUSTRATIONS	
Fig. 1 — 132 kV Wood Pole Design — Outline Detail	33
Fig. 2 — Scarfing and Drilling for Single Intermediate and Section Poles	35
Fig. 3 — Scarfing and Drilling for 'H' Type Intermediate Supports .	36
Fig. 4 — Scarfing and Drilling for Terminal and Section angle 'H' Type Supports	37
Fig. 5 — Scarfing and Drilling — Cable Terminal Structure Auxiliary Pole	38
Fig. 6 — General Arrangement Intermediate Support — Normal Span	39
Fig. 7 — General Arrangement Long Span Intermediate Support Single Pole	41
Fig. 8 — General Arrangement Long Span Intermediate 'H' Pole ...	43
Fig. 9 — General Arrangement of Angle/Section Support Single Pole	45
Fig. 10 — General Arrangement Angle/Section 'H' Pole	47
Fig. 11 — General Arrangement Terminal 'H' Pole	49
Fig. 12 — General Arrangement Pole Mounted Cable Termination ..	51
Fig. 13 — General Arrangement Steel Tower Terminal	53
Fig. 14 — Cable Backboard — Cable Terminal Structure	54
APPENDIX	
A — Lines in Severe Environment	55
Incorporates corrigenda issued April 1985, July 1985 and December 1987	

132 kV SINGLE CIRCUIT OVERHEAD LINES ON WOOD POLES

FOREWORD

The need for single circuit 132 kV lines has encouraged the use of wood poles as an alternative to steel towers, especially in wayleave sensitive areas.

This Standard has therefore been prepared to assist the Electricity Supply Industry in fulfilling this requirement.

1. SCOPE

This Standard covers the design of a three-phase 132 kV single circuit unearthed overhead line employing the following Aluminium Conductor Steel Reinforced (ACSR):

175 mm² (18 + 1/3.61 code name 'Caracal')

175 mm² (30 + 7/2.79 code name 'Lynx')

200 mm² (18 + 1/3.86 code name 'Jaguar')

The design is based on 'Lynx' conductor which imposes the most onerous condition, thus allowing for the erection of any of the above conductors (by reference to the applicable schedules) without substitution of, or amendment to, any of the component parts.

Supports will generally be single wood poles but 'H' pole supports for use on longer spans and heavier angles are included. Provision is made in this Standard for a Maximum Span length of 200 m.

This line design is based on the use of post type insulators of porcelain or reinforced plastic for intermediate positions and normal tension insulators of glass, porcelain or plastic for tension positions.

The overhead line in this Standard is designed to the conditions specified in the Electricity (Overhead Lines) Regulations 1970. Where these conditions may be exceeded due to geographical or climatic conditions, a modified form of construction may be used as described in Appendix A.

2. DEFINITIONS

For the purpose of this Standard the following definitions shall apply:

Intermediate Support — A support in a straight run of a line on which the conductors are supported on line post insulators.

Section Support — A support in a straight run of a line on which the conductors are made off on either side of the crossarm on tension insulator sets.

Section Angle Support — A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets.

Span — The horizontal distance between adjacent supports.

Wind Loading Span — The wind loading span associated with any support is half the sum of the spans adjacent to the support.

Factor of Safety — The ratio, allowed for in the design of the overhead line, between the ultimate stress in a component and the safe permissible stress in it.

Conductor Downpull — The downward vertical load imposed by the conductor at a support resulting from the gradients to the adjacent pole tops.

Conductor Uplift — The upward vertical load imposed by the conductor at a support resulting from the gradients to the adjacent pole tops.

Weight Span — The sum of the distances, either side of a support, to the real or apparent invert points of the catenaries of the adjacent spans.

Basic Span — The span length adopted for sag/tension calculations.

Equivalent Span — The equivalent span of a line, or section of line between rigid points, on which the conductors are free to move longitudinally at the supports, and in which the tension for any condition of load and temperature is therefore the same in all spans, is the span in which the tension variations are identical with those in the line or section of line. Its length is obtained by dividing the sum of the cubes of the individual spans by the length of the line or section and taking the square root of the whole.

i.e.:

$$\text{Equivalent Span} = \sqrt{\frac{L_1^3 + L_2^3 + L_3^3 + \dots}{L_1 + L_2 + L_3 + \dots}}$$

Where L_1 , L_2 and L_3 , are the individual span lengths of the section.

Maximum Span — The maximum permitted length of any span for the conductor spacing adopted.

Sag — The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured at mid-span.

Pretensioning — The tension treatment applied to a conductor before final erection tension is established.

Overtensioning — Excess tension applied above the tension at time of erection.

Auxiliary Equipment — Equipment other than that forming part of the line design which may be placed on supports.

3. REFERENCES

This Standard makes reference to the following documents.

BS 215, 'Aluminium Conductors and Aluminium Conductors, Steel-reinforced, for Overhead Power Transmission'.

Part 2, 'Aluminium Conductors, Steel-reinforced'.

BS 913, 'Wood Preservation by means of Pressure Creosoting'.

BS 1990, 'Wood Poles for Overhead Lines (Power and Telecommunication Lines)'.

BS 3452, 'Copper/Chrome Water-borne Wood Preservatives and their Application'.

BS 3288, 'Insulator and Conductor Fittings for Overhead Power Lines'.

BS 4072, 'Wood Preservation by means of Water-borne Copper/Chrome/Arsenic Compositions'.

*ESI Standard 43—89, 'Auxiliary Equipment Erected on Overhead Line Supports'.
ESI Standard 43—90, 'Anti-climbing Devices for H.V. lines up to and including 400 kV'.
ESI Standard 43—91, 'Stay Strands and Stay Fittings for Overhead Lines'.
ESI Standard 43—92, 'Conductor Terminations, Joints and Insulator Binds for Overhead Lines up to and including 132 kV'.
ESI Standard 43—93, 'Line Insulators'.
ESI Standard 43—95, 'Steelwork for Overhead Lines'.
ESI Standard 43—96, 'Fasteners and Washers for Wood Pole Overhead Lines'.
*ESI Standard 43—8, 'Overhead Line Clearances'.
Engineering Recommendation L.38, 'Protective Greasing of PVC-covered Copper and Bare and PVC-covered Aluminium-based Overhead Line Conductors'.
The Electricity (Overhead Lines) Regulations, 1970, SI 1355.

*In course of preparation.

4. DESIGN DATA

This Standard is designed in accordance with the requirements of the Electricity (Overhead Lines) Regulations 1970.

4.1 Conductors

The minimum factor of safety on conductors shall be 2.0 on their calculated breaking loads as given in BS 215, Part 2.

The maximum design tension of any conductor shall not exceed 23.3 kN.

The design tensions of the conductors at 5°C in still air shall not exceed 20 per cent of their calculated breaking load to reduce the risk of vibration. The conductor mass used in these calculations includes an allowance for grease as specified in Engineering Recommendation L.38 for category 3 minimum allowance plus 1 per cent nominally.

The actual maximum working tensions, factors of safety and conductor mass are tabulated below with the limiting design feature:

ACSR		Conductor Mass kg/m	Max. Working Tension kN	Factor† of Safety	Limiting Feature
Size mm ²	Code Name				
175	Caracal	0.605	17.5	2.34	Vibration Limit
175	Lynx	0.874	23.3	3.42	Tension Limit
200	Jaguar	0.692	19.2	2.42	Vibration Limit

† Factor of Safety is based on 130 m basic span.

4.2 Crossarm Assemblies

The minimum factor of safety on all crossarm assemblies is 2.5 on the ultimate tensile strength of the grade of steel specified.

Crossarm assemblies for intermediate positions are designed to support a conductor downpull on each side of the support, provided that the downpull is acting in a direction such that the tangent of the angle to the horizontal plane is not greater than 0.1.

Crossarm assemblies for section positions are designed to support a conductor downpull on each side of the support, provided that the downpull is acting in a direction such that the tangent of the angle to the horizontal plane is not greater than 0.2.

4.3 Supports

The strength of supports is derived from information contained in BS 1990 and assumes an ultimate extreme fibre stress of 53.8 N/mm^2 .

The bending stresses induced in unstayed supports result from windage on the pole, insulators and acting on the conductor at a point 1.5 m above the pole top.

The wind loading spans are based on pole strengths in accordance with BS 1990 and assume average tapers on pole diameter of 9.25 mm/m for medium poles and 11.25 mm/m for stout poles.

4.4 Insulators and Fittings

A minimum factor of safety of 3 is afforded on 70 kN tension fittings and insulators.

4.5 Foundations

The minimum factor of safety on foundations is 2.5.

Foundation recommendations are designed to match the calculated strength of the poles and provide for ground conditions from normal soil having a compressive strength of $200 \text{ kN/m}^2/\text{m}$ of depth to hard, stoney ground with compressive strength of $400 \text{ kN/m}^2/\text{m}$. For ground below $200 \text{ kN/m}^2/\text{m}$, special load spreading foundations must be employed.

5. MATERIALS

All components used in the construction of lines to this Standard shall comply with British Standards and/or ESI Standards as indicated.

5.1 Conductors

The conductors specified in this Standard shall be in accordance with BS 215, Part 2. Conductors shall be greased in accordance with Engineering Recommendation L.38.

5.2 Wood Poles

Poles shall comply with the requirements of BS 1990. They shall be fabricated to Fig. 2, 3, 4 and 5, all holes being drilled to template, the centre line of the template being placed along the effective centre line of the pole. Pole markings shall be made on the same side of the pole as that in which the drilling templates have been used. Pole tops shall be evenly halved, the angle at the apex being 120° . When fully seasoned and fabricated, poles shall be creosoted in accordance with BS 913 by Rueping Process. An alternative method of wood preservative may be applied in accordance with BS 3452 and BS 4072 if specified by the Purchaser. Where 'H' Poles are required, the limbs shall be suitably matched and marked.

5.3 Crossarm Steelwork

Crossarm steelwork shall comply with ESI Standard 43—95.

5.4 Insulators

Insulators shall comply with ESI Standard 43—93.

5.5 Insulator Fittings

Insulator fittings shall comply with ESI Standard 43—92.

5.6 Conductor Fittings

Conductor terminations, joints and clamps shall comply with ESI Standard 43—92.

5.7 Stays

Stay strand and stay fittings shall comply with ESI Standard 43—91.

5.8 Fasteners and Washers

Fasteners and Washers shall comply with ESI Standard 43—96.

5.9 Danger Plates

Danger plates shall comply with the appropriate Board or CEGB Standard.

6. TECHNICAL REQUIREMENTS FOR THE ERECTION OF LINES

6.1 Span Lengths

This Standard is intended for use with a recommended span of 130 m and a maximum span of 150 m with a minimum mid-span conductor spacing of 2 m. For longer spans, up to a maximum of 200 m, the spacing is increased to 2.5 m and a long intermediate crossarm for use on single or 'H' poles is provided. Section/terminal crossarms are designed with 2.5 m conductor centres to maintain the 2 m mid-span spacing when used with 150 m spans with 75° angles of deviation at the adjacent section positions, or 2.5 m when used as straight line sections or terminals. For mid-span spacings between these limits, maximum span lengths shall be limited to values obtained from Nomogram No. 1.

6.2 Supports

Supports shall be single or 'H' poles assembled in accordance with Fig. 6 to 12.

6.3 Maximum Angles of Deviation

Line deviations are not permitted on post insulators.

Line deviations on single pole section supports are limited to 30°.

Line deviations up to 75° are permitted on 'H' pole section supports, subject to span length and angle limitations described in 6.1.

In extreme situations when line deviations in excess of 75° cannot be avoided, the line should be terminated in each direction with terminal structures in close proximity with suitable jumpers between.

6.4 Wind Loading Spans and Foundations

Recommendations are made in Tables 1 and 2 for the selection of foundation types in relation to ground strength and windspan requirement.

6.5 Capacities of Intermediate Crossarm Assemblies

Loading capacities of crossarm assemblies are based on the use of post type insulators, either plastic or porcelain. The crossarm is designed to accommodate an insulator of maximum weight 140 kg, length 1500 mm, and cantilever strength of 24 kN (MFL) at point of conductor attachment.

If insulators are used having characteristics from the above parameters, the revised vertical load capacities of crossarm assemblies must be calculated.

6.6 Vertical Conductor Loads and Weight Span

The vertical conductor load (V) imposed on a crossarm can be assumed, for practical purposes, to be the sum of the weight of conductor in half the adjacent spans plus the downpull imposed by the gradient on both sides of the support. This may be calculated as follows:

$$V(\text{kg}) = \frac{L_1 + L_2}{2} W + T \max (\tan \theta_1 + \tan \theta_2)$$

$$\text{Weight Span (metres)} = \frac{L_1 + L_2}{2} + \frac{T \max}{W} (\tan \theta_1 + \tan \theta_2)$$

where L_1 and L_2 are adjacent span lengths in metres

W is weight of ice loaded conductor (augmented mass) in kg per metre

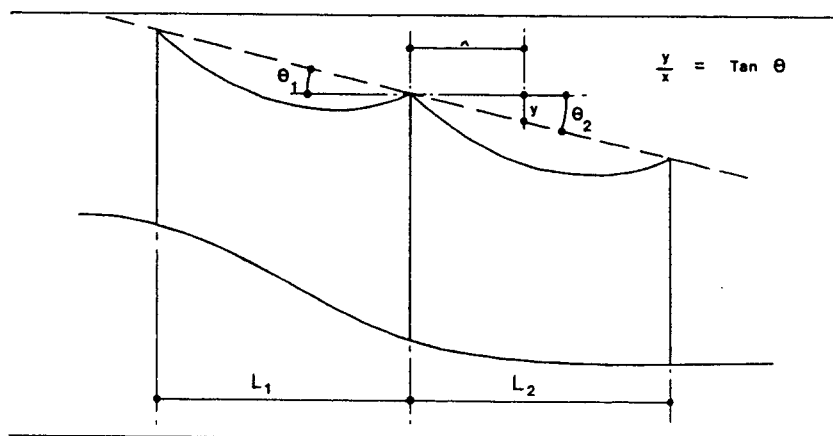
$T \max$ is the maximum design tension for the conductor in kgf

$\tan \theta_1$ and $\tan \theta_2$ are the gradients on each side of the support.

For downpull condition, $\tan \theta$ is positive. For uplift condition, $\tan \theta$ is negative.

On intermediate structures, the gradient on either side of the support, must not exceed 1:10 ($\tan \theta = \pm 0.1$)

On section structures, the gradient on either side must not exceed 1:5 ($\tan \theta = \pm 0.2$)



6.7 Staying Arrangements and Strut Loading for Angle and Terminal Support

Staying requirements will be in accordance with Charts 1 to 5.

Stay assemblies will be in accordance with ESI Standard 43—91.

6.8 Conductor Design and Erection Sags and Tensions

- 6.8.1 If the equivalent span for the total route length of a line does not exceed 130 m, use tables based on 130 m basic span.

If the equivalent span for the total route length of the line exceeds 130 m, use tables based on 170 m basic span.

- 6.8.2 Conductor design sags and tensions for basic spans of 130 m are given in Tables 4 and 5 and for basic spans of 170 m in Tables 9 and 10.

Ordinates for plotting uplift curves and ground clearance curves for basic spans of 130 m are given in Table 3 and for basic spans of 170 m in Table 8.

- 6.8.3 Ordinates for plotting uplift curves are based on conductor design tensions.

Ordinates for plotting ground clearance curves assume the use of a new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20 per cent reduction

30 + 7 ACSR — 5 per cent reduction

- 6.8.4 Conductor pre-tension, erection sags and tensions for basic spans of 130 m are given in Tables 6 and 7 and for basic spans of 170 m in Tables 11 and 12. Erection figures include a value of overtension, equivalent to 10 per cent at 15°C.

Conductors shall be pre-tensioned to the value tabulated in the appropriate pre-tension table for a minimum time of sixty minutes. At 15 minute intervals during this period, the tension shall be readjusted to maintain the pre-tension value.

The conductors shall, at the end of the pre-tension period, be tensioned and terminated in accordance with the appropriate erection table.

6.9 Pilot Insulators

Where pilot insulators are erected on any structure, an equal number per phase shall be used in order to maintain a balanced potential at the crossarm.

To maintain electrical clearance it is recommended that armour rod type conductor clamps are fitted to these insulators.

6.10 Bonding

To avoid a capacitive induced potential, at the point of attachment of insulator support steelwork to the crossarm, it is important that the bases of all three insulators are electrically connected together. This is normally achieved by attachment to the crossarm, but if, for any reason, it is necessary to attach an individual insulator directly to any part of a wood pole, the base of that insulator must be bonded to the bases of the insulators supporting the other two phases.

6.11 Provision of Danger Plates

Each support on a line to this Standard shall be provided with a danger plate in accordance with the appropriate Board or ESI Standard. The plate shall be mounted approximately 3 m above ground level, but above any anti-climbing device.

6.12 Precautions against Access

The types of support requiring precautions against access and the methods to be employed are detailed in ESI Standard 43—90.

6.13 Minimum Height of Line Conductors and Clearances

The minimum height of line conductors above ground shall comply with Electricity (Overhead Lines) Regulations 1970.

Data are given in this Standard for conductors operating at maximum temperatures of 50°C and 75°C.

Clearances to other lines and obstructions shall be in accordance with ESI Standard 43—8 Overhead Line Clearances or other appropriate Board Standard.

6.14 Pole Mounted Cable Termination

A general arrangement for a pole mounted cable termination is given in Fig. 12. All steelwork on this structure shall be bonded together and connected to earth in accordance with ESI 43—89 or other appropriate Standard. Earthing arrangement for cable sealing ends and surge diverters shall be in accordance with the cable system requirements.

6.15 Line Termination on Steel Tower

Where required, the line may be terminated using steel towers as shown in Fig. 13. When a steel tower is used to terminate the line, the adjacent support shall have a conductor spacing of 2.5 m.

6.16 Downloads

The wood pole terminal structure has the facility to accept downloads from a terminal tower or substation structure.

6.17 Download and Jumper Clearances

The following minimum clearances from downloads and jumpers must be maintained and where they are free to move, such movement must be taken into account.

Phase to phase	1.5 m
Phase to support steelwork or wood pole	1.2 m

POLE LENGTH (METRES)

Table 1 --- Windspan/Pole/Foundation Selection Chart --- Single Supports

ENTRY OF 1.5 OR 2.5 UNDER A.B.C. OR D. INDICATES LENGTH OF TOP BAULK

☐ INDICATES FOUNDATION CAPACITY OBTAINED AT LESSER DEPTH OR FOUNDATION TYPE.

ENTRY ☐ INDICATES LOGGATION CAPACITY OBTAINED AT LEVEL DEPTH IN LOGGERS
ENTRY ☒ INDICATES PLANTING DEPTH OR GROUND CLASSIFICATION INSUFFICIENT.

* ANY POLE DIAMETER IN EACH GROUP CAN BE MATCHED WITH ANY PLANTING DEPTH IN THE SAME GROUP.

NOTES ON USE OF TABLE 2

INTERMEDIATE 'H' POLE FOUNDATIONS

Intermediate 'H' Supports should be fitted with wood brace blocks and steel foundation braces (see Fig. 8.) In addition, top baulks, in pairs, can be fitted either 1.5 m or 2.5 m long. The soil conditions, which are such as to necessitate the use of top baulks, are shown in this table.

As indicated in the table, the ground classification is in $\text{kN/m}^2/\text{m}$ of depth.

Classification 100 is equivalent to wet peaty soil, 200 is normal ground, 300 is well drained hard ground and 400 is hard stoney or rocky ground.

Symbols used are as follows:



Indicates planting depth or ground classification insufficient



Indicates foundation possible with 1.5 or 2.5 m top baulks



Indicates foundation possible without top baulks



Indicates foundation possible but capacity could be obtained at lesser depth

* Used in the column between pole diameter and planting depth and indicates that any pole diameter can be matched to any planting depth in the horizontal groups which are bracketed.

Where the ground is permanently saturated and buoyancy is expected, increase the indicated planting depth by one increment.

POLE LENGTH (m)	WINDSPAN REQUIREMENT (m)							POLE DIA. (mm)	DEPTH (m)	GROUND CLASSIFICATION					
	50	75	100	125	150	175	200			100	200	300	400		
		10						*	230	2.0	0				
		10.5	10						235						
		11	10.5						240	2.25					
		11.5	11						245						
		12	11.5	10				*	250	2.0		1.5	0		
			12	10.5					255						
		13		11					260	2.25	1.5	0			
			13	11.5	10				265						
				12	10.5				270	2.5	0				
			14		11	10			275						
				13	11.5	10.5			280	2.75					
					12	11			285						
			15	14		11.5	10		290						
						12	10.5		295	3.0					
				15	13/14		11		*	300	2.0			2.5	0
				16		13	11.5			305					
					15		12	310		2.25		1.5	0		
						14		315							
				17	16		13	320		2.5	1.5	0			
						15	14	325							
				18	17			330		2.75	0				
						16	15	335							
						18		340		3.0					
						17	16	345							
						18	17	*		350	2.0				1.5
										355					
									360	2.25					
									365			2.5	1.5		
									370	2.5	2.5	0			
									375						
							380		2.75	0					
							385								
							390	3.0							
							395								
							*	400	2.5	2.5	0				
								405							
								410	2.75	0					
								415							
								420	3.0						
								425							

WINDSPAN CAPACITIES FOR MINIMUM BS 1990 POLE SIZES												
D \ L	10	10.5	11	11.5	12	13	14	15	16	17	18	
230	96											
235		96										
240			97									
245				97								
250					97							
260						98						
275							107					
285	198											
290		198						116				
295			198									
300				198								
305					198				125			
320						200				134		
330											135	
335							200					
350								200				
365									200			
375										200		
390											200	

POLE LENGTH (m)	BASIC BURYING DEPTH (m)
UP TO 15 MEDIUM	2.25
UP TO 11 STOUT	2.25
12 - 18 STOUT	2.5

GROUND CLASSIFICATION
IS IN kN/m²/m

FACTOR OF SAFETY 2.5

Table 2 — Windspan/Pole/Foundation Selection Chart — 'H' Supports

**Table 3 — Ordinates for Plotting Uplift and Ground Clearance Curves —
Basic Span 130 m**

Span (m) (2X)	Sag (m)					
	175 mm ² 30 + 7/2.79 mm Lynx			175 mm ² 18 + 1/3.61 mm Caracal 200 mm ² 18 + 1/3.86 mm Jaguar		
	-6°C (Still Air)	50°C	75°C	-6°C (Still Air)	50°C	75°C
40	0.11	0.22	0.26	0.12	0.28	0.33
80	0.43	0.87	1.05	0.48	1.13	1.31
120	0.98	1.97	2.36	1.09	2.54	2.95
160	1.74	3.49	4.20	1.94	4.51	5.23
200	2.72	5.46	6.56	3.02	7.05	8.18
240	3.91	7.86	9.44	4.37	10.15	11.78
280	5.33	10.70	12.86	5.93	13.82	16.03
320	6.95	13.97	16.79	7.76	18.05	20.94
360	8.80	17.69	21.25	9.80	22.84	26.50
400	10.87	21.83	26.23	12.13	28.20	32.72
440	13.15	26.42	31.74	14.64	34.12	39.59
480	15.65	31.44	37.78	17.46	40.61	47.11
520	18.37	36.90	44.34	20.44	47.66	55.30
560	21.30	42.79	51.42	23.77	55.27	64.13
600	24.45	49.13	59.03	27.21	63.45	73.62
640	27.82	55.89	67.16	31.05	72.19	83.76
680	31.41	63.10	75.82	34.95	81.50	94.56

Note: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20 per cent reduction
30 + 7 ACSR — 5 per cent reduction

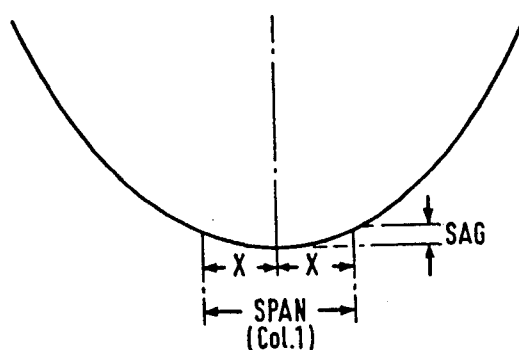


Table 4 — Design Sag Tension Table for 175 mm² (30 + 7/2.79) ACSR 'Lynx' — Basic Span 130 m

Temp (°C)	Tension (kgf)	Sag (m) for Span Lengths (m)																	
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0	
0.0	1481	0.12	0.18	0.27	0.36	0.47	0.60	0.74	0.89	1.06	1.25	1.45	1.66	1.89	2.13	2.39	2.66	2.95	
5.0	1383	0.13	0.20	0.28	0.39	0.51	0.64	0.79	0.96	1.14	1.34	1.55	1.78	2.02	2.28	2.56	2.85	3.16	
10.0	1293	0.14	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.22	1.43	1.66	1.90	2.16	2.44	2.74	3.05	3.38	
15.0	1211	0.14	0.23	0.32	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03	2.31	2.61	2.92	3.26	3.61	
20.0	1138	0.15	0.24	0.35	0.47	0.61	0.78	0.96	1.16	1.38	1.62	1.88	2.16	2.46	2.78	3.11	3.47	3.84	
25.0	1071	0.16	0.25	0.37	0.50	0.65	0.83	1.02	1.23	1.47	1.72	2.00	2.29	2.61	2.95	3.30	3.68	4.08	
30.0	1012	0.17	0.27	0.39	0.53	0.69	0.87	1.08	1.31	1.55	1.82	2.12	2.43	2.76	3.12	3.50	3.90	4.32	
35.0	959	0.18	0.28	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.93	2.23	2.56	2.92	3.29	3.69	4.11	4.56	
40.0	911	0.19	0.30	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89	4.33	4.80	
45.0	868	0.20	0.31	0.45	0.62	0.81	1.02	1.26	1.52	1.81	2.13	2.47	2.83	3.22	3.64	4.08	4.55	5.04	
50.0	829	0.21	0.33	0.47	0.65	0.84	1.07	1.32	1.59	1.90	2.23	2.58	2.96	3.37	3.81	4.27	4.76	5.27	
55.0	794	0.22	0.34	0.50	0.67	0.88	1.11	1.38	1.66	1.98	2.32	2.70	3.09	3.52	3.97	4.46	4.97	5.51	
60.0	763	0.23	0.36	0.52	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.81	3.22	3.67	4.14	4.64	5.17	5.73	
65.0	734	0.24	0.37	0.54	0.73	0.95	1.21	1.49	1.80	2.14	2.51	2.92	3.35	3.81	4.30	4.82	5.37	5.95	
70.0	708	0.25	0.39	0.56	0.76	0.99	1.25	1.54	1.87	2.22	2.61	3.02	3.47	3.95	4.46	5.00	5.57	6.17	
75.0	684	0.26	0.40	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13	3.59	4.09	4.61	5.17	5.76	6.38	

Table 5 — Design Sag Tension Table for 175 mm² (18 + 1/3.61) and 200 mm² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' — Basic Span 130 m

Temp (°C)	Tension (kgf)		Sag (m) for Span Lengths (m)																
	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0
0.0	905	1028	0.13	0.21	0.30	0.41	0.53	0.68	0.83	1.01	1.20	1.41	1.63	1.88	2.13	2.41	2.70	3.01	3.33
5.0	836	950	0.14	0.23	0.33	0.44	0.58	0.73	0.90	1.09	1.30	1.52	1.77	2.03	2.31	2.61	2.92	3.25	3.61
10.0	776	882	0.16	0.24	0.35	0.48	0.62	0.79	0.97	1.18	1.40	1.64	1.91	2.19	2.49	2.81	3.15	3.51	3.89
15.0	723	822	0.17	0.26	0.38	0.51	0.67	0.85	1.04	1.26	1.50	1.77	2.05	2.35	2.67	3.02	3.38	3.77	4.17
20.0	676	769	0.18	0.28	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.89	2.19	2.51	2.86	3.23	3.62	4.03	4.47
25.0	635	723	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33	2.67	3.04	3.43	3.85	4.29	4.75
30.0	599	683	0.20	0.32	0.45	0.62	0.81	1.01	1.26	1.52	1.81	2.13	2.47	2.83	3.22	3.64	4.08	4.55	5.04
35.0	568	647	0.21	0.33	0.48	0.65	0.85	1.08	1.33	1.61	1.91	2.25	2.61	2.99	3.40	3.84	4.31	4.80	5.32
40.0	540	615	0.22	0.35	0.50	0.69	0.90	1.13	1.40	1.69	2.01	2.36	2.74	3.15	3.58	4.04	4.53	5.07	5.60
45.0	515	587	0.23	0.37	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	2.87	3.30	3.75	4.24	4.75	5.29	5.86
50.0	493	562	0.25	0.38	0.55	0.75	0.98	1.24	1.53	1.85	2.21	2.59	3.00	3.45	3.92	4.43	4.96	5.53	6.12
55.0	473	540	0.26	0.40	0.58	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13	3.59	4.09	4.61	5.17	5.76	6.38
60.0	455	519	0.27	0.42	0.60	0.81	1.06	1.34	1.66	2.01	2.39	2.80	3.25	3.73	4.25	4.80	5.38	5.99	6.64
65.0	439	501	0.28	0.43	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87	4.40	4.97	5.57	6.21	6.88
70.0	424	484	0.29	0.45	0.64	0.87	1.14	1.44	1.78	2.15	2.56	3.01	3.49	4.01	4.56	5.15	5.77	6.43	7.12
75.0	411	469	0.30	0.46	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.11	3.61	4.14	4.71	5.32	5.96	6.64	7.36

Table 6 — Pretension and Erection Table for 175 mm² (30 + 7/2.79) ACSR ‘Lynx’ — Basic Span 130 m

Pretension kgf (all Temps)	Sag (m) for Span Lengths (m)																
	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200
1800	0.09	0.15	0.22	0.29	0.38	0.48	0.60	0.73	0.86	1.01	1.17	1.34	1.53	1.73	1.93	2.15	2.38

Erection Temp (°C)	Erection Tension (kgf)	Sag (m) for Span Lengths (m)																
		40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200
0	1637	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.96	1.13	1.31	1.50	1.71	1.93	2.16	2.41	2.67
5	1528	0.11	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.21	1.40	1.61	1.83	2.07	2.32	2.58	2.86
10	1426	0.12	0.19	0.28	0.38	0.49	0.62	0.77	0.93	1.10	1.29	1.50	1.72	1.96	2.21	2.48	2.76	3.06
15	1333	0.13	0.20	0.30	0.40	0.52	0.66	0.82	0.99	1.18	1.39	1.61	1.84	2.10	2.37	2.66	2.96	3.28
20	1247	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.26	1.48	1.72	1.97	2.24	2.53	2.84	3.16	3.51
25	1170	0.15	0.23	0.34	0.46	0.60	0.76	0.93	1.13	1.34	1.58	1.83	2.10	2.39	2.70	3.03	3.38	3.74
30	1100	0.16	0.25	0.36	0.49	0.64	0.80	0.99	1.20	1.43	1.68	1.95	2.23	2.54	2.87	3.22	3.59	3.98
35	1038	0.17	0.26	0.38	0.52	0.67	0.85	1.05	1.27	1.52	1.78	2.06	2.37	2.69	3.04	3.41	3.80	4.21
40	982	0.18	0.28	0.40	0.55	0.71	0.90	1.11	1.35	1.60	1.88	2.18	2.50	2.85	3.22	3.61	4.02	4.46
45	932	0.19	0.29	0.42	0.57	0.75	0.95	1.17	1.42	1.69	1.98	2.30	2.64	3.00	3.39	3.80	4.23	4.69
50	887	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.42	2.77	3.15	3.56	3.99	4.45	4.93
55	846	0.21	0.32	0.46	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91	3.31	3.73	4.18	4.66	5.16
60	810	0.22	0.34	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.64	3.04	3.45	3.90	4.37	4.87	5.40
65	777	0.23	0.35	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.16	3.60	4.07	4.56	5.08	5.63
70	747	0.23	0.37	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	2.87	3.29	3.75	4.23	4.74	5.28	5.85
75	720	0.24	0.38	0.55	0.74	0.97	1.23	1.52	1.84	2.19	2.57	2.98	3.42	3.89	4.39	4.92	5.48	6.07

Table 7 — Pretension and Erection Table for 175 mm² (18 + 1/3.61) and 200 mm² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' —
Basic Span 130 m

Pretension kgf (all Temps)		Sag (m) for Span Lengths (m)																
175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0
950	1050	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63	2.93	3.25

Erection Temp (°C)	Erection Tension (kgf)		Sag (m) for Span Lengths (m)																
	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0
			0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	
0	1009	1146	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.91	1.08	1.26	1.47	1.68	1.92	2.16	2.42	2.70	2.99
5	929	1056	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63	2.93	3.25
10	858	976	0.14	0.22	0.32	0.43	0.56	0.71	0.88	1.06	1.27	1.49	1.72	1.98	2.25	2.54	2.85	3.18	3.52
15	795	904	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.60	1.86	2.14	2.43	2.74	3.08	3.43	3.80
20	739	841	0.16	0.26	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	2.00	2.30	2.61	2.95	3.31	3.69	4.09
25	691	786	0.18	0.27	0.39	0.54	0.70	0.89	1.09	1.32	1.57	1.85	2.14	2.46	2.80	3.16	3.54	3.94	4.37
30	648	738	0.19	0.29	0.42	0.57	0.75	0.94	1.17	1.41	1.68	1.97	2.28	2.62	2.98	3.37	3.77	4.20	4.65
35	611	696	0.20	0.31	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42	2.78	3.17	3.57	4.01	4.47	4.95
40	578	659	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.58	1.88	2.21	2.56	2.94	3.35	3.78	4.23	4.71	5.22
45	549	626	0.22	0.34	0.50	0.67	0.88	1.11	1.38	1.67	1.98	2.33	2.70	3.10	3.52	3.98	4.46	4.97	5.51
50	523	597	0.23	0.36	0.52	0.71	0.92	1.17	1.44	1.75	2.08	2.44	2.83	3.25	3.70	4.17	4.68	5.21	5.78
55	500	570	0.24	0.38	0.54	0.74	0.97	1.22	1.51	1.83	2.18	2.55	2.96	3.40	3.87	4.37	4.89	5.45	6.04
60	480	547	0.25	0.39	0.57	0.77	1.00	1.28	1.58	1.91	2.27	2.66	3.09	3.54	4.03	4.55	5.10	5.68	6.30
65	461	526	0.26	0.41	0.59	0.80	1.05	1.33	1.64	1.98	2.36	2.77	3.21	3.69	4.20	4.74	5.31	5.92	6.56
70	445	507	0.27	0.43	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.87	3.33	3.83	4.35	4.92	5.51	6.14	6.80
75	429	490	0.28	0.44	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.45	3.96	4.51	5.09	5.71	6.36	7.05

**Table 8 — Ordinates for Plotting Uplift and Ground Clearance Curves —
Basic Span 170 m**

Span (m) (2X)	Sag (m)					
	175 mm ² 30 + 7/2.79 mm Lynx			175 mm ² 18 + 1/3.61 mm Caracal 200 mm ² 18 + 1/3.86 mm Jaguar		
	- 6°C (Still Air)	50°C	75°C	- 6°C (Still Air)	50°C	75°C
40	0.13	0.21	0.24	0.13	0.25	0.28
80	0.52	0.84	0.96	0.51	1.00	1.12
120	1.17	1.89	2.16	1.15	2.25	2.52
160	2.08	3.37	3.83	2.03	3.99	4.48
200	3.25	5.27	5.59	3.17	6.23	7.00
240	4.68	7.59	8.62	4.57	8.97	10.08
280	6.37	10.33	11.73	6.22	12.21	13.72
320	8.31	13.49	15.33	8.13	15.95	17.91
360	10.52	17.07	19.40	10.29	20.19	22.67
400	12.99	21.07	23.95	12.71	24.92	27.99
440	15.72	25.49	28.98	15.38	30.15	33.87
480	18.71	30.34	34.49	18.30	35.88	40.31
520	21.96	35.61	40.48	21.48	42.11	47.31
560	25.46	41.30	46.95	24.90	48.84	54.86
600	29.23	47.41	53.90	28.58	56.07	62.98
640	33.26	53.94	61.32	32.53	63.79	71.66
680	37.55	60.89	69.22	36.72	72.01	80.90

Note: Ordinates for plotting uplift curves are based on conductor design tensions. Ordinates for plotting ground clearance curves assume the use of new conductor and include an allowance for long term creep over a period of 25 years. This allowance is equivalent to reductions of the design tension values at a temperature of 15°C as indicated below:

18 + 1 ACSR — 20 per cent reduction
30 + 7 ACSR — 5 per cent reduction

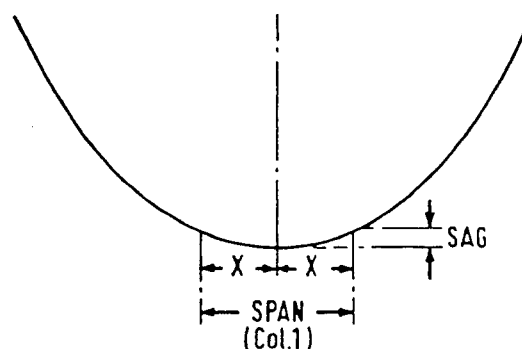


Table 9 — Design Sag Tension Table for 175 mm² (30 + 7/2.79) ACSR 'Lynx' — Basic Span 170 m

Temp (°C)	Tension (kgf)	Sag (m) for Span Lengths (m)																
		40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0
0	1267	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94	2.21	2.49	2.79	3.11	3.45
5	1209	0.15	0.23	0.33	0.44	0.58	0.73	0.90	1.09	1.30	1.53	1.77	2.03	2.31	2.61	2.93	3.26	3.62
10	1155	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.14	1.36	1.60	1.85	2.13	2.42	2.73	3.07	3.42	3.78
15	1106	0.16	0.25	0.36	0.48	0.63	0.80	0.99	1.20	1.42	1.67	1.94	2.22	2.53	2.86	3.20	3.57	3.95
20	1061	0.17	0.26	0.37	0.50	0.66	0.83	1.03	1.25	1.48	1.74	2.02	2.32	2.64	2.98	3.34	3.72	4.12
25	1020	0.17	0.27	0.39	0.52	0.69	0.87	1.07	1.30	1.54	1.81	2.10	2.41	2.74	3.09	3.47	3.87	4.28
30	983	0.18	0.28	0.40	0.54	0.71	0.90	1.11	1.35	1.60	1.88	2.18	2.50	2.85	3.21	3.60	4.01	4.45
35	948	0.19	0.29	0.41	0.56	0.74	0.93	1.15	1.39	1.66	1.95	2.26	2.59	2.95	3.33	3.73	4.16	4.61
40	917	0.19	0.30	0.43	0.58	0.76	0.97	1.19	1.44	1.72	2.01	2.34	2.68	3.05	3.44	3.86	4.30	4.77
45	888	0.20	0.31	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.06	2.41	2.77	3.15	3.56	3.99	4.44	4.92
50	861	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.54	1.83	2.15	2.49	2.86	3.25	3.67	4.11	4.58	5.08
55	836	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.58	1.88	2.21	2.56	2.94	3.35	3.78	4.24	4.72	5.23
60	812	0.22	0.34	0.48	0.66	0.86	1.09	1.34	1.63	1.94	2.27	2.64	3.03	3.44	3.89	4.36	4.86	5.38
65	791	0.22	0.35	0.50	0.68	0.88	1.12	1.38	1.67	1.99	2.34	2.71	3.11	3.54	3.99	4.48	4.99	5.53
70	771	0.23	0.35	0.51	0.69	0.91	1.15	1.42	1.72	2.04	2.40	2.78	3.19	3.63	4.10	4.59	5.12	5.67
75	752	0.23	0.36	0.52	0.71	0.93	1.18	1.45	1.76	2.09	2.46	2.85	3.27	3.72	4.20	4.71	5.25	5.81

Table 10 — Design Sag Tension Table for 175 mm² (18 + 1/3.61) and 200 mm² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' — Basic Span 170 m

Temp (°C)	Tension (kgf)		Sag (m) for Span Lengths (m)																
	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0
0	886	1006	0.13	0.21	0.31	0.42	0.55	0.69	0.85	1.03	1.23	1.44	1.67	1.92	2.19	2.47	2.77	3.08	3.42
5	836	950	0.14	0.23	0.33	0.44	0.58	0.73	0.90	1.09	1.30	1.53	1.77	2.03	2.32	2.61	2.93	3.27	3.62
10	792	901	0.15	0.24	0.34	0.47	0.61	0.77	0.95	1.16	1.38	1.61	1.87	2.15	2.44	2.76	3.09	3.45	3.82
15	753	856	0.16	0.25	0.36	0.49	0.64	0.81	1.01	1.22	1.45	1.70	1.97	2.26	2.57	2.90	3.26	3.63	4.02
20	717	816	0.17	0.26	0.38	0.52	0.68	0.85	1.05	1.28	1.52	1.78	2.07	2.37	2.70	3.05	3.42	3.81	4.22
25	685	780	0.18	0.28	0.40	0.54	0.71	0.89	1.10	1.34	1.59	1.87	2.16	2.48	2.83	3.19	3.58	3.98	4.42
30	656	747	0.19	0.29	0.41	0.56	0.74	0.93	1.15	1.39	1.66	1.95	2.26	2.59	2.95	3.33	3.73	4.16	4.61
35	630	718	0.19	0.30	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	2.35	2.70	3.07	3.47	3.89	4.33	4.80
40	607	691	0.20	0.31	0.45	0.61	0.80	1.01	1.25	1.51	1.79	2.11	2.44	2.80	3.19	3.60	4.04	4.50	4.99
45	585	667	0.20	0.32	0.47	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53	2.91	3.31	3.73	4.19	4.66	5.17
50	566	645	0.21	0.33	0.48	0.66	0.86	1.08	1.34	1.62	1.93	2.26	2.62	3.01	3.42	3.86	4.33	4.83	5.35
55	548	624	0.22	0.35	0.50	0.68	0.88	1.12	1.38	1.67	1.99	2.33	2.71	3.11	3.54	3.99	4.47	4.99	5.52
60	531	606	0.23	0.36	0.51	0.70	0.91	1.15	1.42	1.72	2.05	2.41	2.79	3.20	3.65	4.12	4.61	5.14	5.70
65	516	588	0.24	0.37	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	2.87	3.30	3.75	4.24	4.75	5.29	5.87
70	502	572	0.24	0.38	0.54	0.74	0.97	1.22	1.51	1.82	2.17	2.55	2.96	3.39	3.86	4.36	4.89	5.44	6.03
75	488	557	0.25	0.39	0.56	0.76	0.99	1.25	1.55	1.87	2.23	2.62	3.04	3.48	3.96	4.48	5.02	5.59	6.20

Table 11 — Pretension and Erection Table for 175 mm² (30 + 7/2.79) ACSR 'Lynx' — Basic Span 170 m

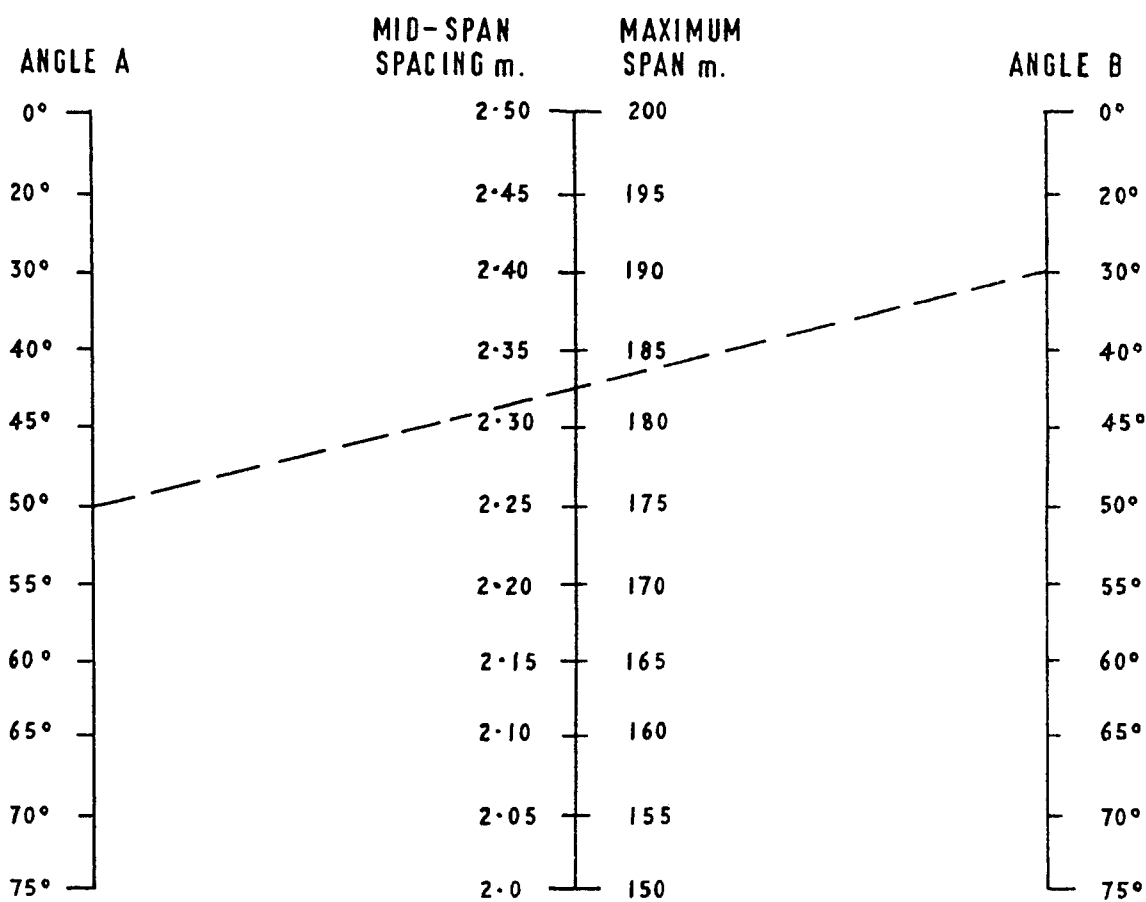
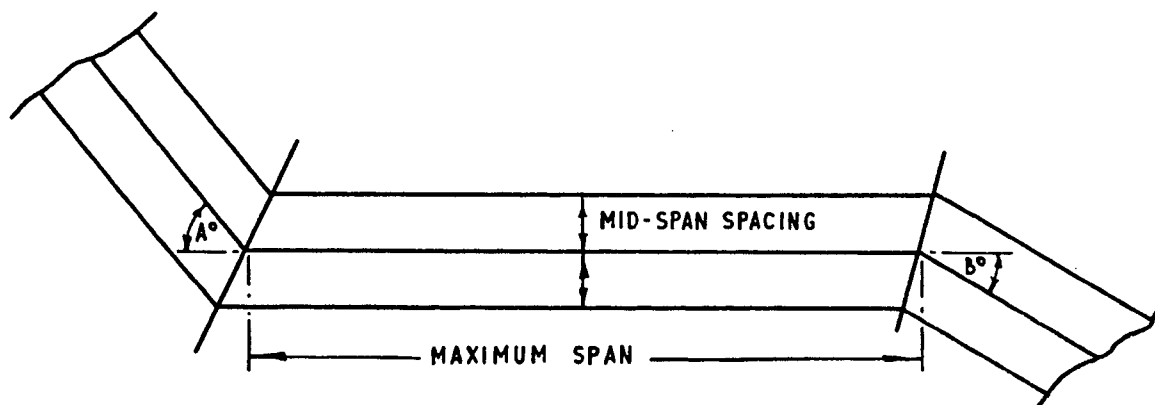
Pretension kgf (all Temps)	Sag (m) for Span Lengths (m)																
	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200
1800	0.09	0.15	0.22	0.29	0.38	0.48	0.60	0.73	0.86	1.01	1.17	1.34	1.53	1.73	1.93	2.15	2.38

Temp (°C)	Tension (kgf)	Sag (m) for Span Lengths (m)																
		40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200
0	1413	0.12	0.19	0.28	0.38	0.49	0.63	0.77	0.94	1.11	1.31	1.52	1.74	1.98	2.23	2.51	2.79	3.09
5	1341	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.99	1.17	1.38	1.60	1.83	2.09	2.35	2.64	2.94	3.26
10	1276	0.13	0.21	0.31	0.42	0.55	0.69	0.86	1.04	1.23	1.45	1.68	1.93	2.19	2.47	2.77	3.09	3.42
15	1217	0.14	0.22	0.32	0.44	0.57	0.73	0.90	1.09	1.29	1.52	1.76	2.02	2.30	2.60	2.91	3.24	3.59
20	1162	0.15	0.24	0.34	0.46	0.60	0.76	0.94	1.14	1.35	1.59	1.84	2.12	2.41	2.72	3.05	3.39	3.76
25	1113	0.15	0.25	0.35	0.48	0.63	0.80	0.98	1.19	1.41	1.66	1.92	2.21	2.51	2.84	3.18	3.55	3.93
30	1067	0.16	0.26	0.37	0.50	0.66	0.83	1.02	1.24	1.47	1.73	2.01	2.30	2.62	2.96	3.32	3.70	4.09
35	1026	0.17	0.27	0.38	0.52	0.68	0.86	1.07	1.29	1.53	1.80	2.09	2.40	2.73	3.08	3.45	3.84	4.26
40	988	0.17	0.28	0.40	0.54	0.71	0.90	1.11	1.34	1.59	1.87	2.17	2.49	2.83	3.20	3.58	3.99	4.42
45	953	0.18	0.29	0.41	0.55	0.73	0.93	1.15	1.39	1.65	1.94	2.25	2.58	2.93	3.31	3.71	4.14	4.59
50	921	0.19	0.30	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.00	2.33	2.67	3.04	3.43	3.84	4.28	4.74
55	891	0.19	0.31	0.44	0.60	0.78	0.99	1.23	1.48	1.76	2.07	2.40	2.76	3.14	3.54	3.97	4.42	4.90
60	864	0.20	0.32	0.46	0.62	0.81	1.02	1.26	1.53	1.82	2.14	2.48	2.84	3.24	3.65	4.10	4.56	5.06
65	839	0.21	0.33	0.47	0.64	0.83	1.05	1.30	1.58	1.88	2.20	2.55	2.93	3.33	3.76	4.22	4.70	5.21
70	816	0.21	0.33	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.26	2.63	3.01	3.43	3.87	4.34	4.84	5.36
75	794	0.22	0.34	0.50	0.67	0.88	1.12	1.38	1.67	1.98	2.33	2.70	3.10	3.52	3.98	4.46	4.97	5.51

Table 12 — Pretension and Erection Table for 175 mm² (18 + 1/3.61) and 200 mm² (18 + 1/3.86) ACSR 'Caracal' and 'Jaguar' —
Basic Span 170 m

Pretension kgf (all Temps)		Sag (m) for Span Lengths (m)																
175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0
950	1050	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63	2.93	3.25

Erection Temp (°C)	Erection Tension (kgf)		Sag (m) for Span Lengths (m)																
	175 mm ²	200 mm ²	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0	170.0	180.0	190.0	200.0
0	990	1125	0.12	0.19	0.27	0.37	0.49	0.62	0.76	0.92	1.10	1.29	1.50	1.72	1.96	2.21	2.47	2.76	3.05
5	930	1057	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	1.59	1.83	2.08	2.35	2.63	2.94	3.25
10	876	996	0.14	0.22	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	1.69	1.94	2.21	2.49	2.80	3.12	3.45
15	828	941	0.15	0.23	0.33	0.45	0.58	0.74	0.91	1.11	1.32	1.54	1.79	2.06	2.34	2.64	2.96	3.30	3.65
20	784	893	0.15	0.24	0.35	0.47	0.62	0.78	0.96	1.17	1.39	1.63	1.89	2.17	2.47	2.79	3.12	3.48	3.86
25	746	849	0.16	0.25	0.37	0.50	0.65	0.82	1.01	1.23	1.46	1.71	1.99	2.28	2.60	2.93	3.29	3.66	4.06
30	711	810	0.17	0.27	0.38	0.52	0.68	0.86	1.06	1.29	1.53	1.80	2.09	2.39	2.72	3.07	3.45	3.84	4.26
35	680	774	0.18	0.28	0.40	0.55	0.71	0.90	1.11	1.35	1.60	1.88	2.18	2.50	2.85	3.22	3.61	4.02	4.45
40	651	742	0.19	0.29	0.42	0.57	0.74	0.94	1.16	1.40	1.67	1.96	2.28	2.61	2.97	3.36	3.76	4.19	4.65
45	626	713	0.19	0.30	0.43	0.59	0.77	0.98	1.21	1.46	1.74	2.04	2.37	2.72	3.09	3.49	3.91	4.31	4.83
50	603	687	0.20	0.31	0.45	0.61	0.80	1.02	1.25	1.52	1.81	2.12	2.46	2.82	3.21	3.63	4.07	4.53	5.02
55	582	663	0.21	0.33	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.20	2.55	2.93	3.33	3.76	4.21	4.69	5.20
60	562	641	0.22	0.34	0.48	0.66	0.86	1.09	1.35	1.63	1.94	2.27	2.64	3.03	3.44	3.89	4.36	4.86	5.38
65	544	621	0.22	0.35	0.50	0.68	0.89	1.13	1.39	1.68	2.00	2.35	2.72	3.13	3.56	4.01	4.50	5.01	5.58
70	528	603	0.23	0.36	0.52	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.81	3.22	3.67	4.14	4.64	5.17	5.73
75	513	585	0.24	0.37	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.49	2.89	3.32	3.77	4.26	4.78	5.32	5.90



[E.G. FOR ANGLES 'A' = 50°, 'B' = 30°
MID-SPAN SPACING IS 2.325m AND MAXIMUM SPAN IS 182.5m]

Nomogram No. 1
Mid-span Spacing/Maximum Span Relationship
Adjacent Angles

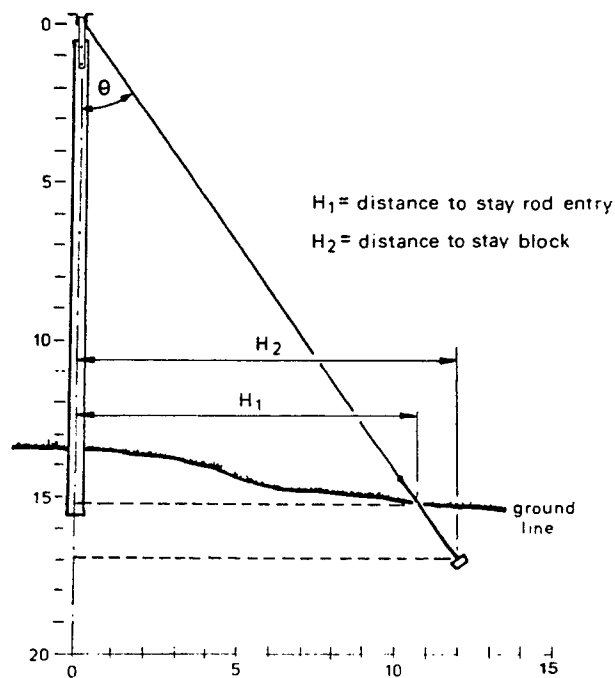
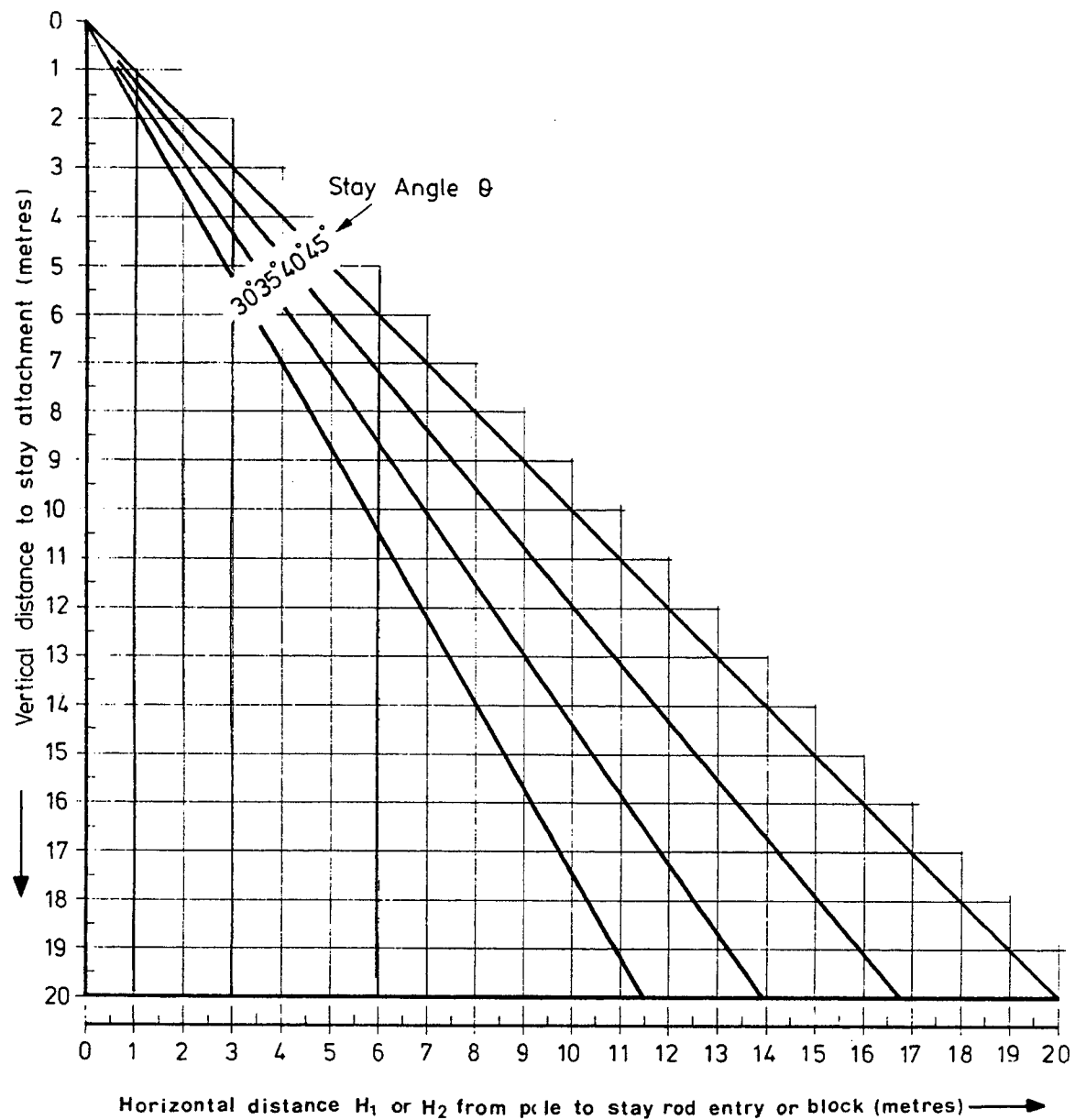


Chart 1 — Stay Angles

(2) Wind load on projected area of support of 380 newtons per square metre on windward poles and 190 newtons per square metre on leeward poles.

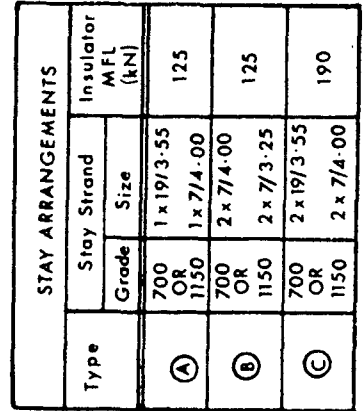
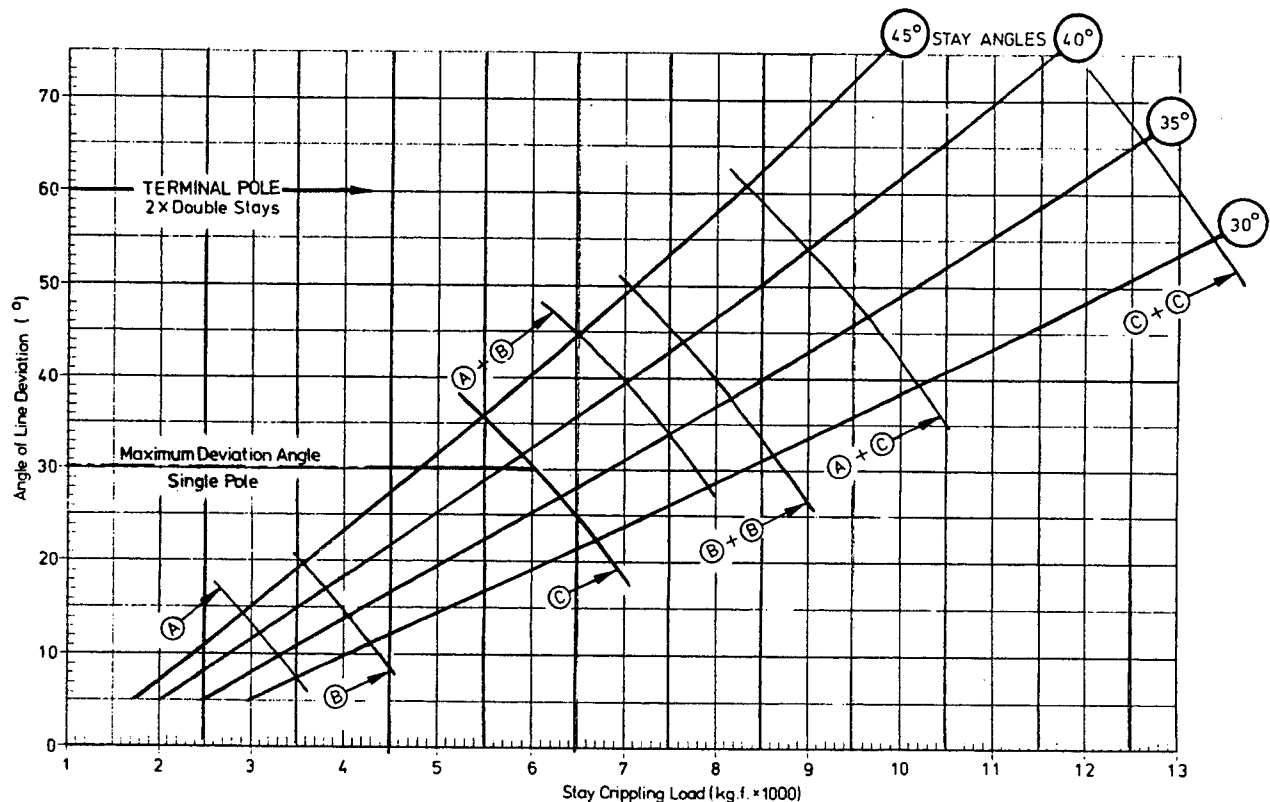


Chart 2 — Stayed Pole Designs



STAY ARRANGEMENTS			
Type	Stay Strand		Insulator MFL (kN)
	Grade	Size	
A	700	1x19/3.55	125
	OR 1150	1x7/4.00	
B	700	2x7/4.00	125
	OR 1150	2x7/3.25	
C	700	2x19/3.55	190
	OR 1150	2x7/4.00	

CONDUCTOR DOWNLOADS		
Gradient	Downpull (kgf x 1000)	
	H Poles	Single Poles
Level Ground	0.8	1.6
1:30	1.0	2.1
1:20	1.1	2.3
1:10	1.5	3.0

DESIGN BASIS

Conductors

- (1) Wind loading span 150 m, bare conductor.
- (2) Maximum working tension 23.3 kN.
- (3) Wind load of 380 N/m² per square metre acting on projected area of conductor augmented by 19 mm.

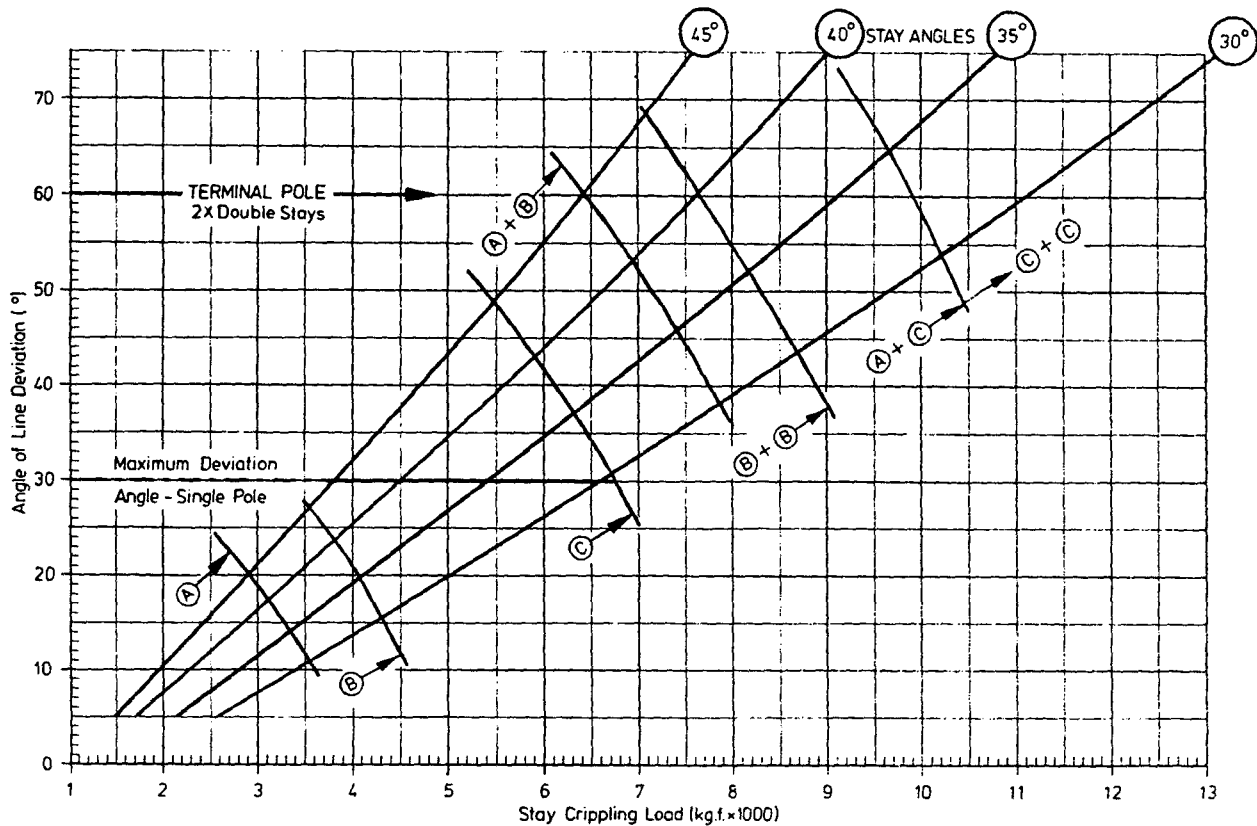
Stays

- (1) In accordance with ESI Standard 43—91.

To use this chart:

- (1) From angle of line deviation take horizontal line to selected stay angle. Use 60° deviation for terminal pole.
- (2) Point obtained from (1) gives minimum stay size for various stay assemblies as indicated by arrows on stay curves.
- (3) From point obtained in (1) take vertical line to obtain Stay Crippling Load — note value.
- (4) Add appropriate Conductor Download from table to Stay Crippling Load obtained from (3). Also add the weight of any additional pole mounted equipment.
- (5) Load obtained from (4) is the Total Strut Loading on Support and can be used to determine pole size required for individual members by reference to Chart 2 (Stayed Pole Design Chart).

Chart 3 — Stay Sizes and Strut Loads for 175 mm² (30 + 7/2.79) ACSR 'Lynx'



STAY ARRANGEMENTS			
Type	Stay Strand		Insulator MFL (kN)
	Grade	Size	
A	700	1x19/3.55	125
	OR 1150	1x7/4.00	
B	700	2x7/4.00	125
	OR 1150	2x7/3.25	
C	700	2x19/3.55	190
	OR 1150	2x7/4.00	

CONDUCTOR DOWNLOADS		
Gradient	Downpull (kgf.×1000)	
	H Poles	Single Poles
Level Ground	0.7	1.4
1:30	0.9	1.8
1:20	1.0	2.0
1:10	1.2	2.5

DESIGN BASIS

Conductors

- (1) Wind loading span 150 m, bare conductor.
- (2) Maximum working tension 17.5 kN.
- (3) Wind load of 380 N/m² per square metre acting on projected area of conductor augmented by 19 mm.

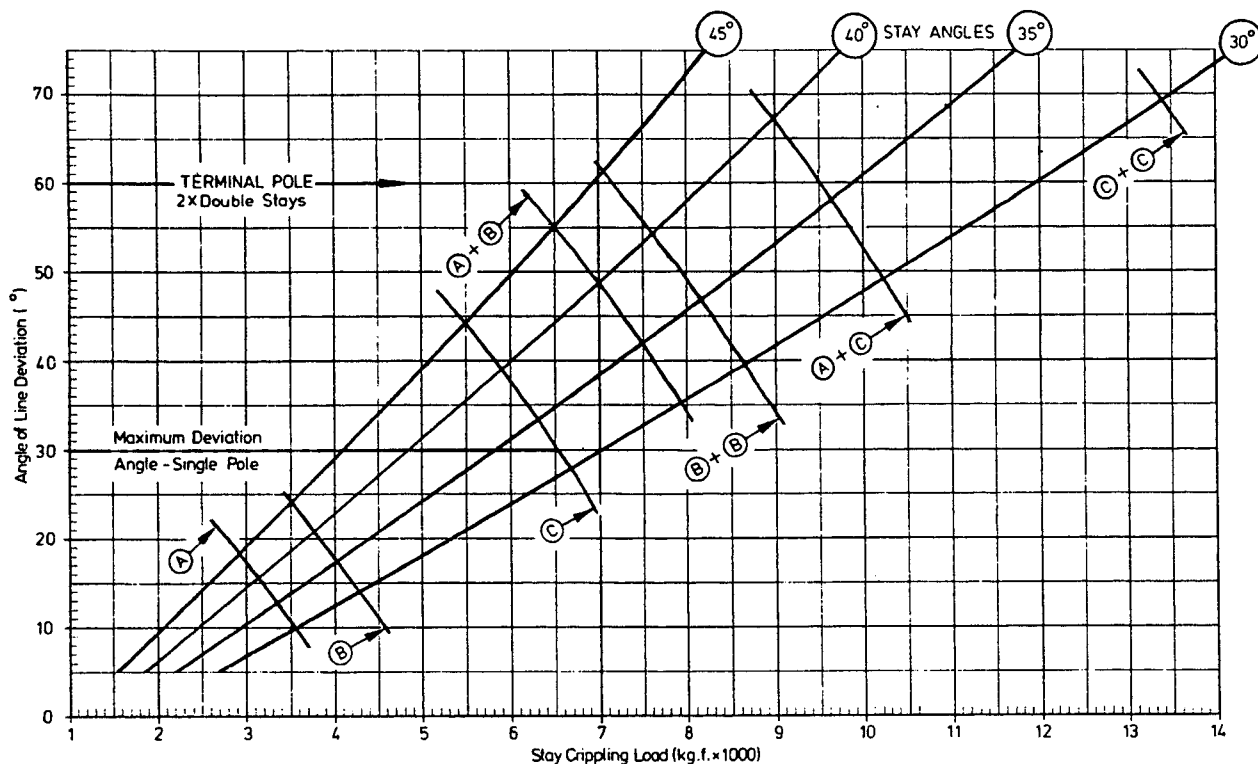
Stays

- (1) In accordance with ESI Standard 43—91.

To use this chart:

- (1) From angle of line deviation take horizontal line to selected stay angle. Use 60° deviation for terminal pole.
- (2) Point obtained from (1) gives minimum stay size for various stay assemblies as indicated by arrows on stay curves.
- (3) From point obtained in (1) take vertical line to obtain Stay Crippling Load — note value.
- (4) Add appropriate Conductor Download from table to Stay Crippling Load obtained from (3). Also add the weight of any additional pole mounted equipment.
- (5) Load obtained from (4) is the Total Strut Loading on Support and can be used to determine pole size required for individual members by reference to Chart 2 (Stayed Pole Design Chart).

Chart 4 — Stay Sizes and Strut Loads for 175 mm² (18 + 1/3.61) ACSR 'Caracal'



STAY ARRANGEMENTS			
Type	Stay Strand		Insulator MFL (kN)
	Grade	Size	
A	700	1x 19/3.55	125
	OR 1150	1x 7/4.00	
B	700	2x 7/4.00	125
	OR 1150	2x 7/3.25	
C	700	2x 19/3.55	190
	OR 1150	2x 7/4.00	

CONDUCTOR DOWNLOADS		
Gradient	Downpull (kgf x 1000)	
	H Poles	Single Poles
Level Ground	0.7	1.5
1:30	0.9	1.9
1:20	1.0	2.1
1:10	1.3	2.7

DESIGN BASIS**Conductors**

- (1) Wind loading span 150 m, bare conductor.
- (2) Maximum working tension 19.2 kN.
- (3) Wind load of 380 N/m² per square metre acting on projected area of conductor augmented by 19 mm.

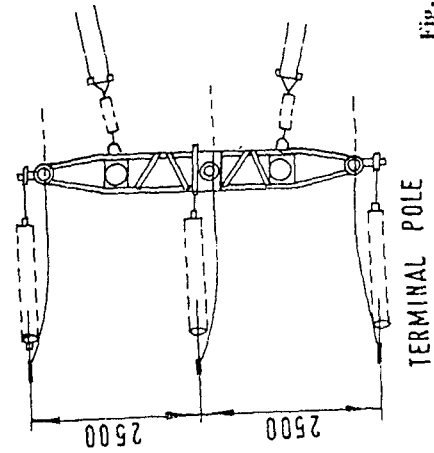
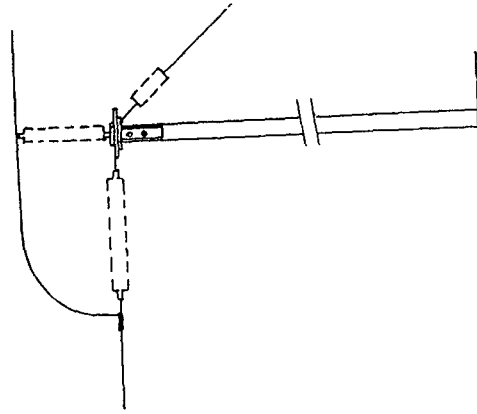
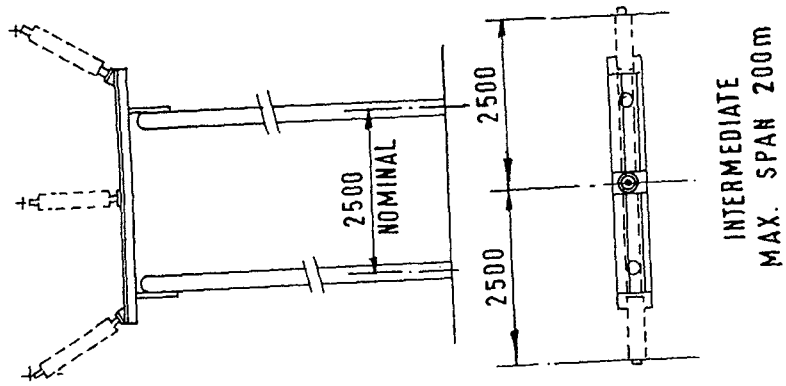
Stays

- (1) In accordance with ESI Standard 43—91.

To use this chart:

- (1) From angle of line deviation take horizontal line to selected stay angle. Use 60° deviation for terminal pole.
- (2) Point obtained from (1) gives minimum stay size for various stay assemblies as indicated by arrows on stay curves.
- (3) From point obtained in (1) take vertical line to obtain Stay Crippling Load — note value.
- (4) Add appropriate Conductor Download from table to Stay Crippling Load obtained from (3). Also add the weight of any additional pole mounted equipment.
- (5) Load obtained from (4) is the Total Strut Loading on Support and can be used to determine pole size required for individual members by reference to Chart 2 (Stayed Pole Design Chart).

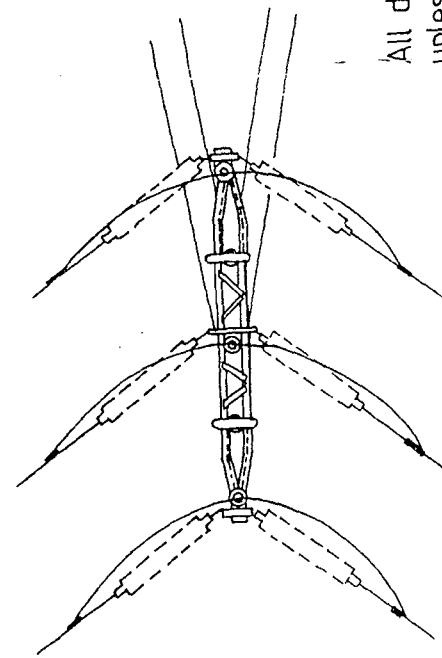
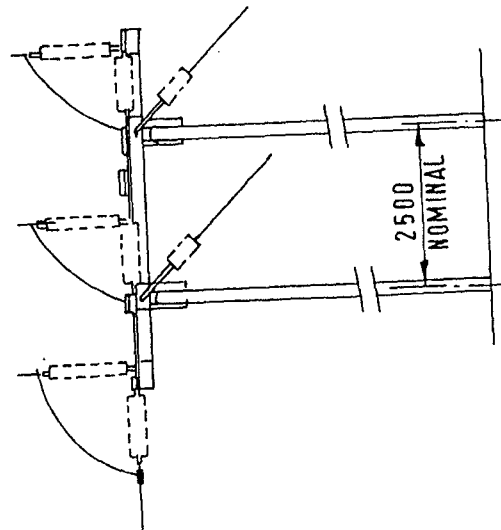
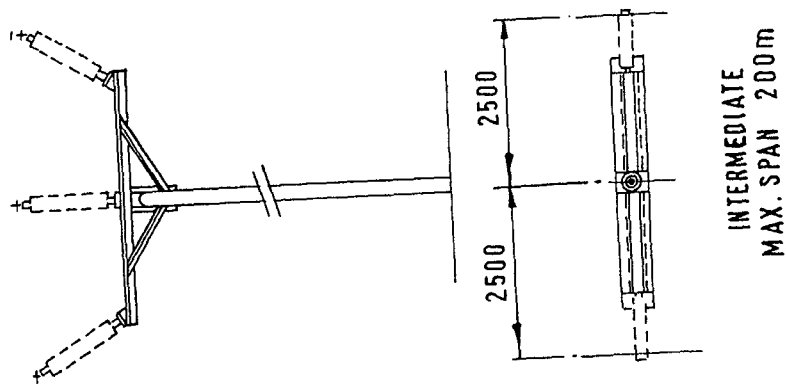
Chart 5 — Stay Sizes and Strut Loads for 200 mm² (18 + 1/3.86) ACSR 'Jaguar'



68/6620 Rev. A

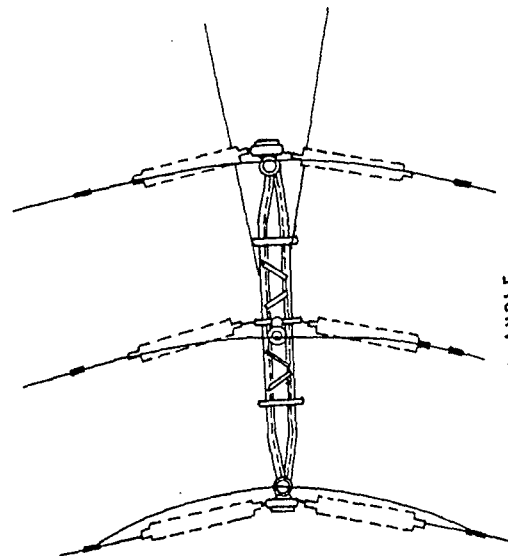
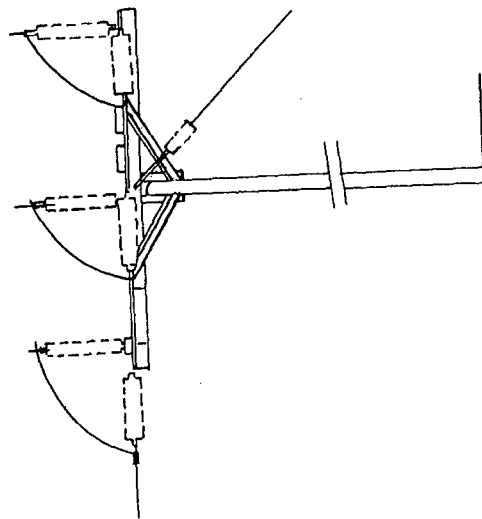
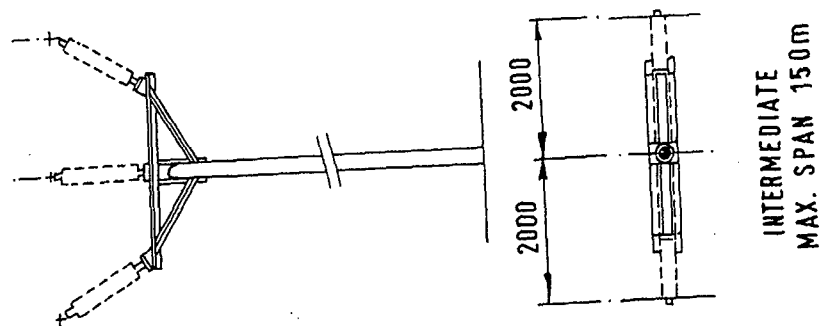
ESI Drawing No. 435001

Fig. 1 — 132 kV Wood Pole Design —
Outline Detail

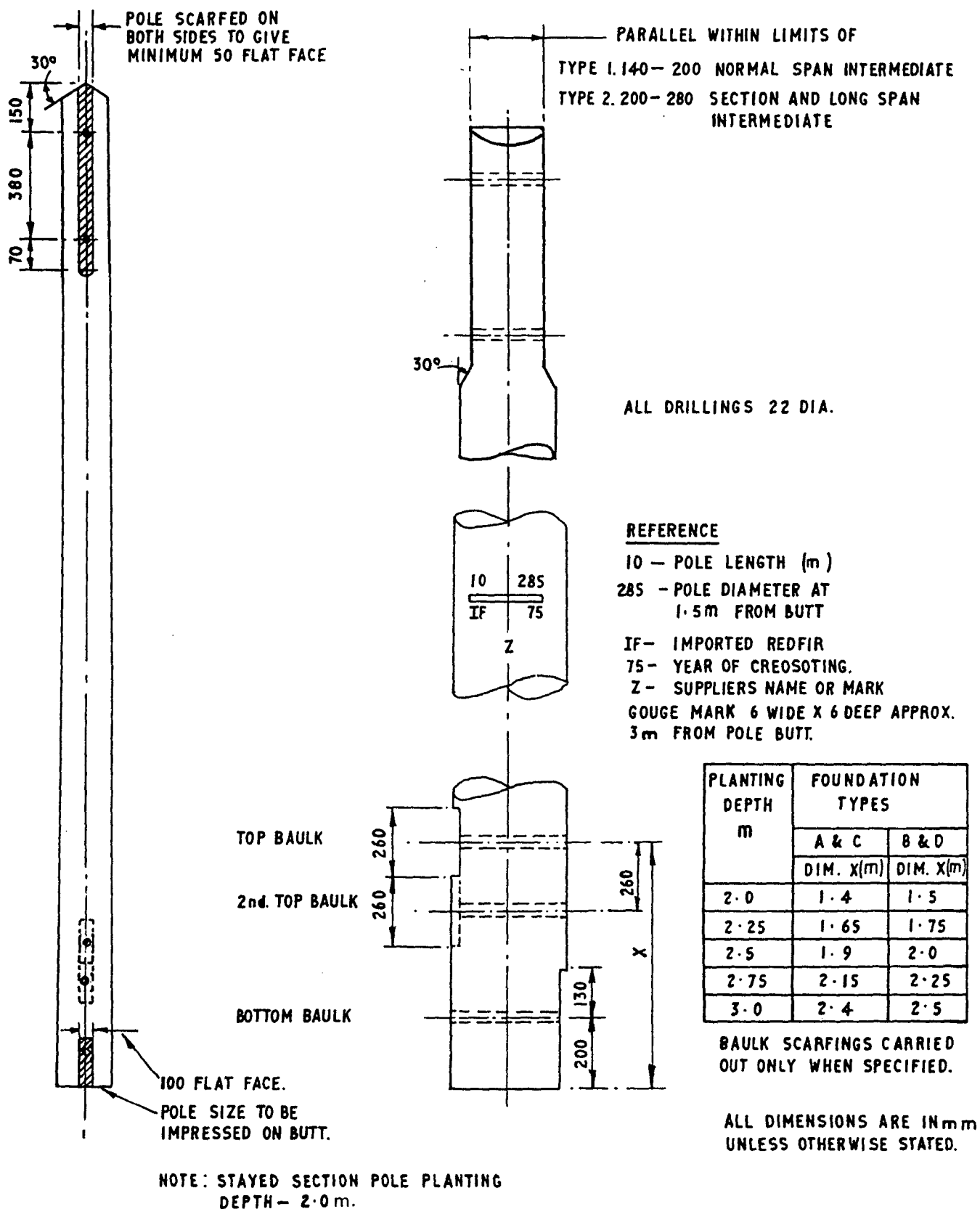


All dimensions in mm
unless otherwise stated

SECTION ANGLE
MAX. DEVIATION 75°

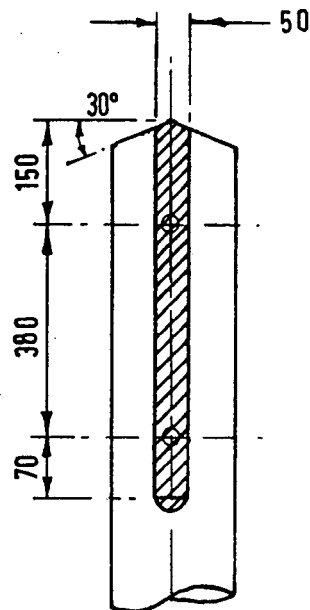
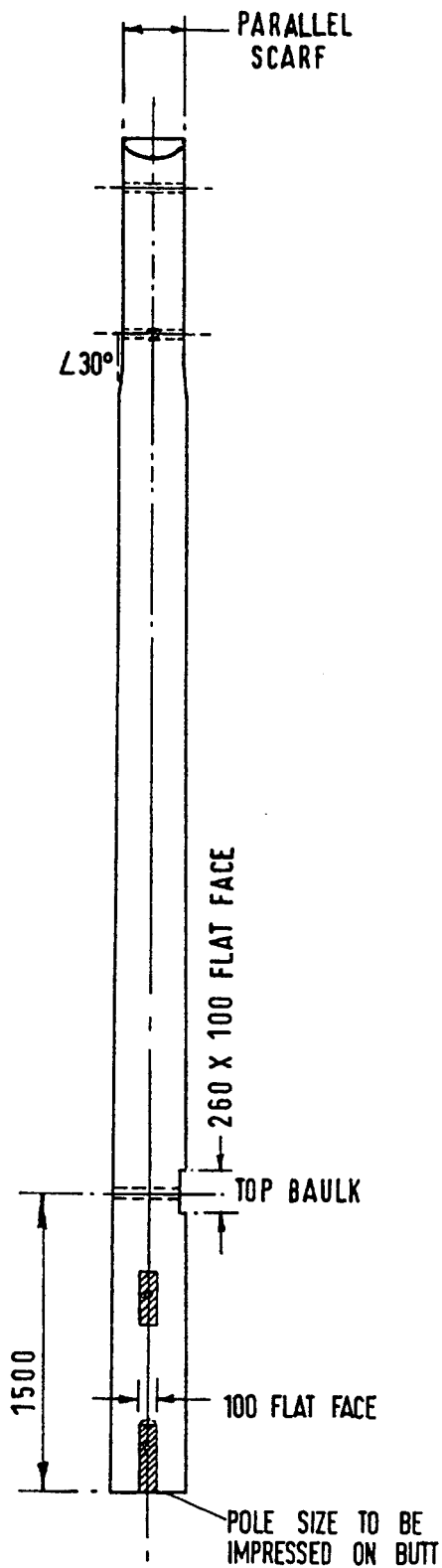


SECTION ANGLE
MAX. DEVIATION 30°



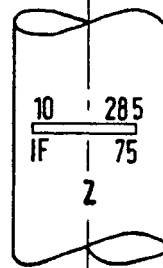
ESI Drawing No. 435002

Fig. 2 — Scarfing and drilling for single intermediate and section poles



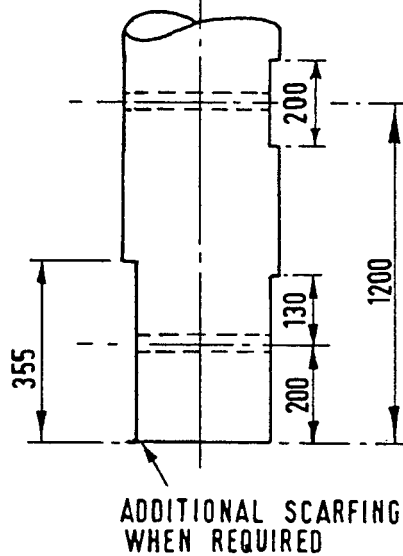
POLE SCARFED ON BOTH SIDES TO GIVE 50 FLAT FACE.

ALL DRILLINGS 22 DIA.



REFERENCE

10 - POLE LENGTH (m)
 285 POLE DIAMETER AT 1.5m FROM BUTT
 IF - IMPORTED REDFIR
 75 - YEAR OF CREOSOTING
 Z - SUPPLIERS NAME OR MARK
 GOUGE MARK 6 WIDE X 6 DEEP 3m FROM POLE BUTT



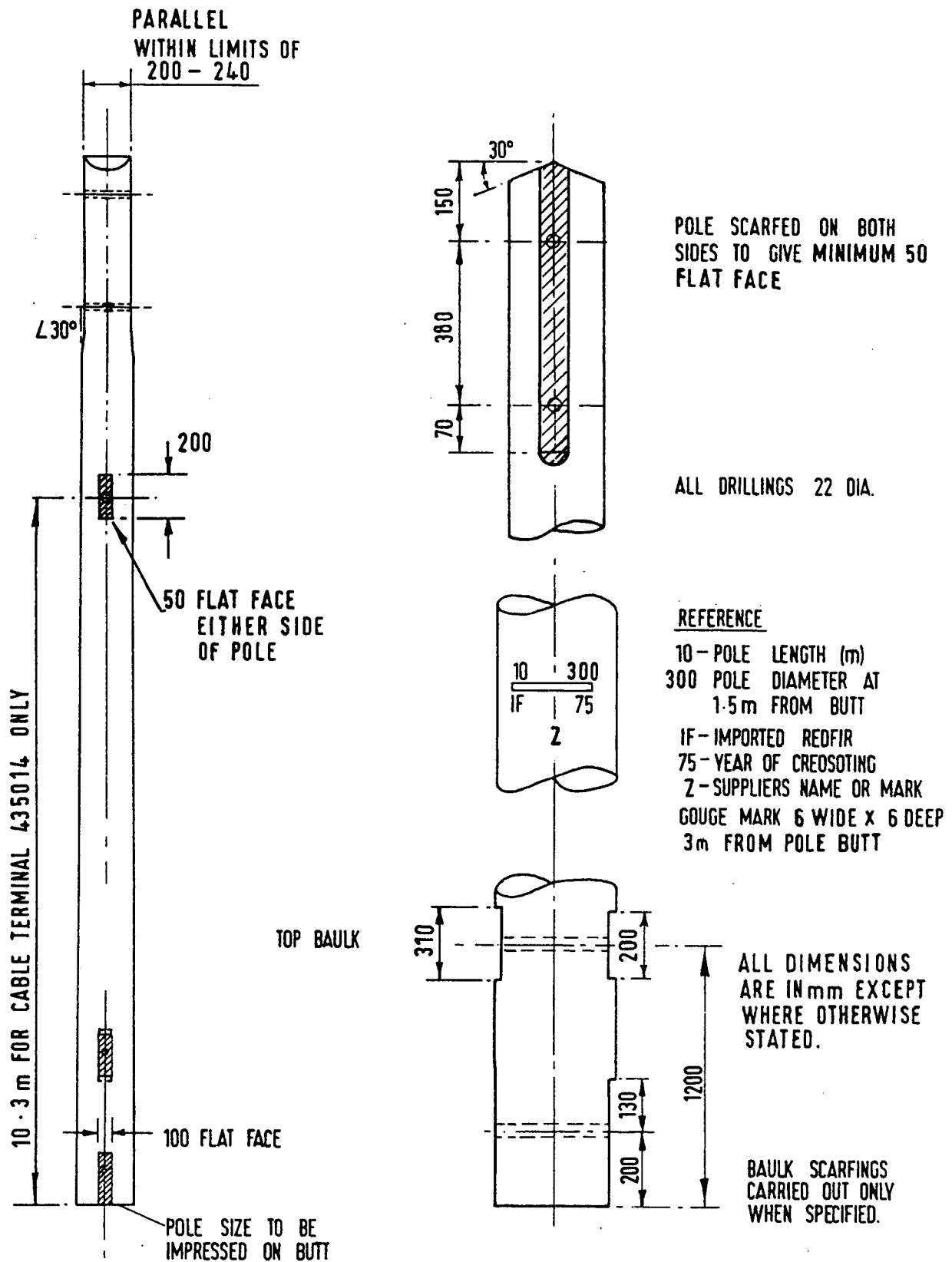
ALL DIMENSIONS ARE IN mm EXCEPT WHERE OTHERWISE STATED.

BAULK SCARFINGS CARRIED OUT ONLY WHEN SPECIFIED.

ESI Drawing No. 435003

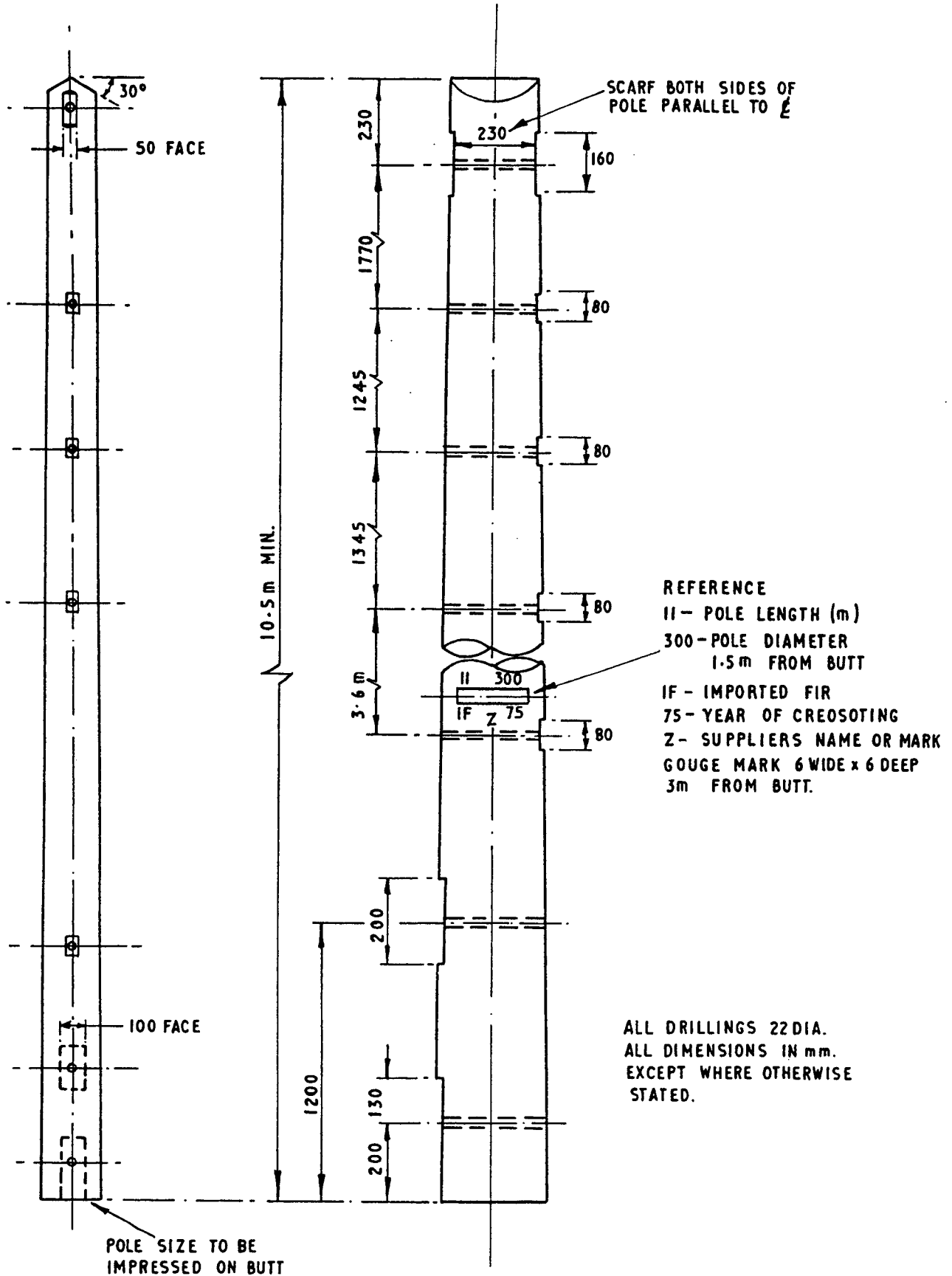
Fig. 3 — Scarfing and drilling 'H' type intermediate supports

68/6622



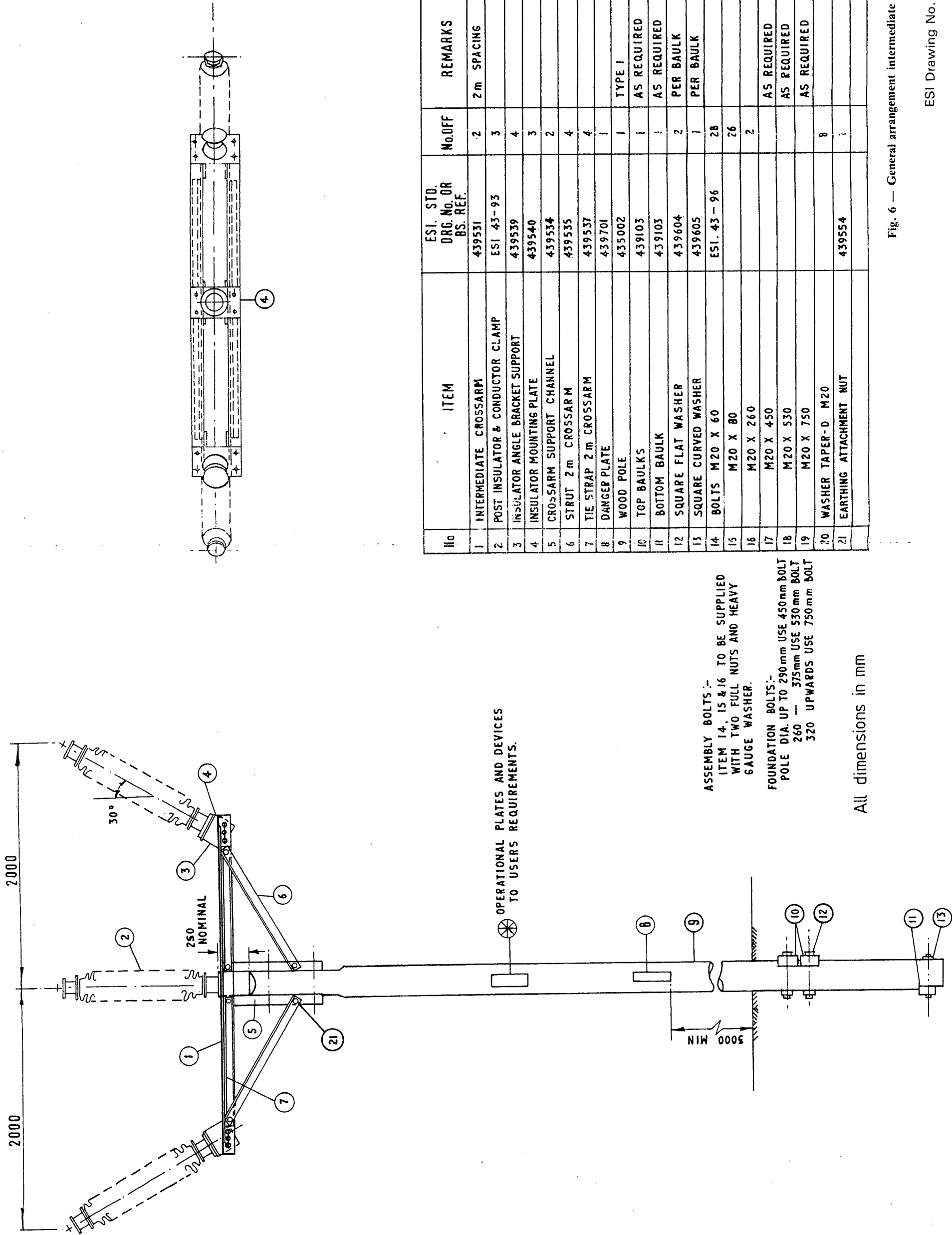
ESI Drawing No. 435004

Fig. 4 — Scarfing and drilling for terminal and section angle 'H' type supports



ESI Drawing No. 435005

Fig. 5 — Scarfing and drilling — cable terminal structure auxiliary pole



No	ITEM	ESI. STD. DRG. No. OR BS. REF.	No. OFF	REMARKS
1	INTERMEDIATE CROSSARM	439531	2	2 m SPACING
2	POST INSULATOR & CONDUCTOR CLAMP	ESI 43-93	3	
3	INSULATOR ANGLE BRACKET SUPPORT	439539	4	
4	INSULATOR MOUNTING PLATE	439540	3	
5	CROSSARM SUPPORT CHANNEL	439534	2	
6	STRUT 2 m CROSSARM	439535	4	
7	TIE STRAP 2 m CROSSARM	439537	4	
8	DANGER PLATE	439701	1	
9	WOOD POLE	435002	1	TYPE 1
10	TOP BAULKS	439103	1	AS REQUIRED
11	BOTTOM BAULK	439103	1	AS REQUIRED
12	SQUARE FLAT WASHER	439604	2	PER BAULK
13	SQUARE CURVED WASHER	439605	1	PER BAULK
14	BOLTS M20 X 60	ESI. 43-96	28	
15	M20 X 80		26	
16	M20 X 260		2	
17	M20 X 450			AS REQUIRED
18	M20 X 530			AS REQUIRED
19	M20 X 750			AS REQUIRED
20	WASHER TAPER-D M20		8	
21	EARTHING ATTACHMENT NUT	439554	1	

Fig. 6 — General arrangement intermediate support — normal span

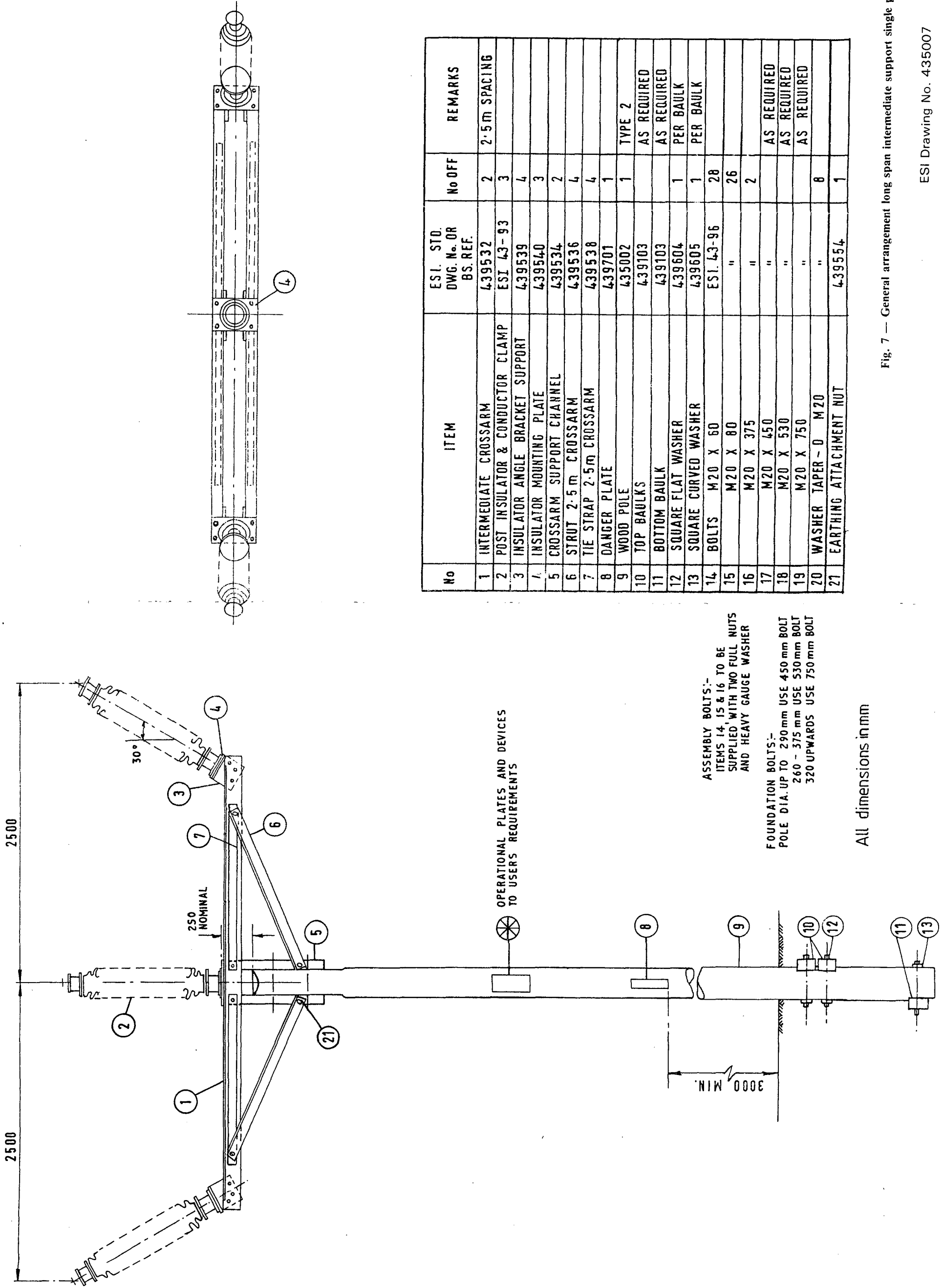
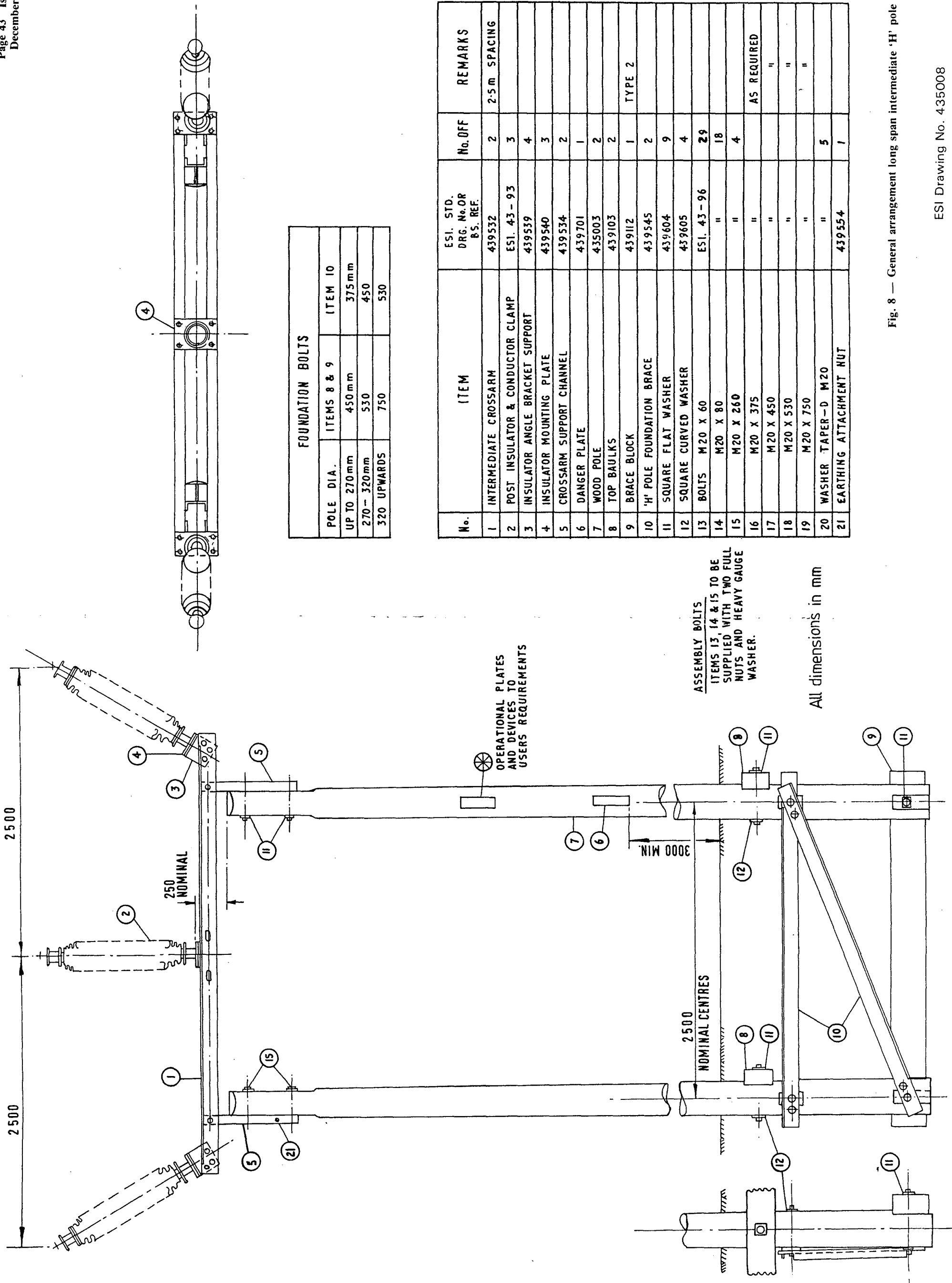


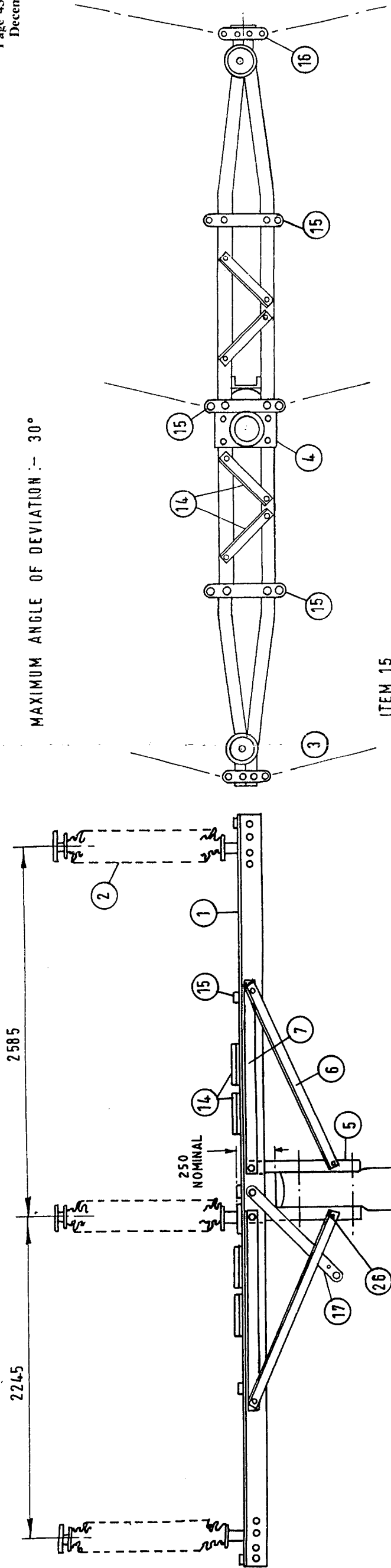
Fig. 7 — General arrangement long span intermediate support single pole



FOUNDATION BOLTS		
POLE DIA.	ITEMS 8 & 9	ITEM 10
UP TO 270mm	450 mm	375 mm
270 - 320mm	530	450
320 UPWARDS	750	530

No.	ITEM	ESI. STD. DRG. No. OR B.S. REF.	No. OFF	REMARKS
1	INTERMEDIATE CROSSARM	439532	2	2.5 m SPACING
2	POST INSULATOR & CONDUCTOR CLAMP	ESI. 43 - 93	3	
3	INSULATOR ANGLE BRACKET SUPPORT	439539	4	
4	INSULATOR MOUNTING PLATE	439540	3	
5	CROSSARM SUPPORT CHANNEL	439534	2	
6	DANGER PLATE	439701	1	
7	WOOD POLE	435003	2	
8	TOP BAULKS	439103	2	
9	BRACE BLOCK	439112	1	TYPE 2
10	'H' POLE FOUNDATION BRACE	439545	2	
11	SQUARE FLAT WASHER	439604	9	
12	SQUARE CURVED WASHER	439605	4	
13	BOLTS M20 X 60	ESI. 43 - 96	29	
14	M20 X 80	"	18	
15	M20 X 260	"	4	
16	M20 X 375	"		AS REQUIRED
17	M20 X 450	"		"
18	M20 X 530	"		"
19	M20 X 750	"		"
20	WASHER TAPER-D M20	"	5	
21	EARTHING ATTACHMENT NUT	439554	1	

Fig. 8 — General arrangement long span intermediate 'H' pole



ITEM 15
OUTER LONG TYPE STRAPS ARE FITTED TO BRACE THE CROSSARM AND MUST REMAIN IN POSITION. MAY BE USED FOR STAY ATTACHMENT DURING CONSTRUCTION.

No.	ITEM	ESI. STD. DWG. No. OR BS. REF.	No. OFF	REMARKS
1	ANGLE CROSSARM	439533	2	
2	POST INSULATOR & CONDUCTOR CLAMP	ESI. 43-93	3	
3	TENSION INSULATORS AND	ESI. 43-93	6-SETS	
4	INSULATOR FITTINGS	BS. 3288	6-SETS	
5	INSULATOR MOUNTING PLATE	439540	1	
6	CROSSARM SUPPORT CHANNEL	439534	2	
7	STRUT 2.5m CROSSARM	439536	4	
8	TIE STRAP 2.5m CROSSARM	439538	4	
9	DANGER PLATE	439701	1	
10	WOOD POLE	435002	1	TYPE 2
11	TOP BAULKS	439103		AS REQUIRED
12	BOTTOM BAULK	439103		AS REQUIRED
13	SQUARE FLAT WASHER	439604	1	PER BAULK
14	SQUARE CURVED WASHER	439605	1	PER BAULK
15	CROSS BRACING	439543	4	
16	LONG TYPE STRAPS	439542	3	
17	SHORT TYPE STRAPS	439541	2	
18	STAY ASSEMBLY	ESI. 43-91		
19	BOLTS M20 X 60	ESI. 43-96	34	
20	M20 X 80	"	8	
21	M24 X 60	"	2	
22	M20 X 375	"	2	
23	M20 X 450	"		AS REQUIRED
24	M20 X 530	"		AS REQUIRED
25	M20 X 750	"		AS REQUIRED
26	WASHER TAPER 'O'	"	8	
27	EARTHING ATTACHMENT NUT	439554	1	

ASSEMBLY BOLTS:-
ITEMS 18, 19, 20 & 21 TO BE
SUPPLIED WITH TWO FULL NUTS
AND HEAVY GAUGE WASHER.

FOUNDATION BOLTS:-
POLE DIA. UP TO 290mm USE 450mm BOLT
260 - 375mm USE 530mm BOLT
320 - UPWARD USE 750mm BOLT

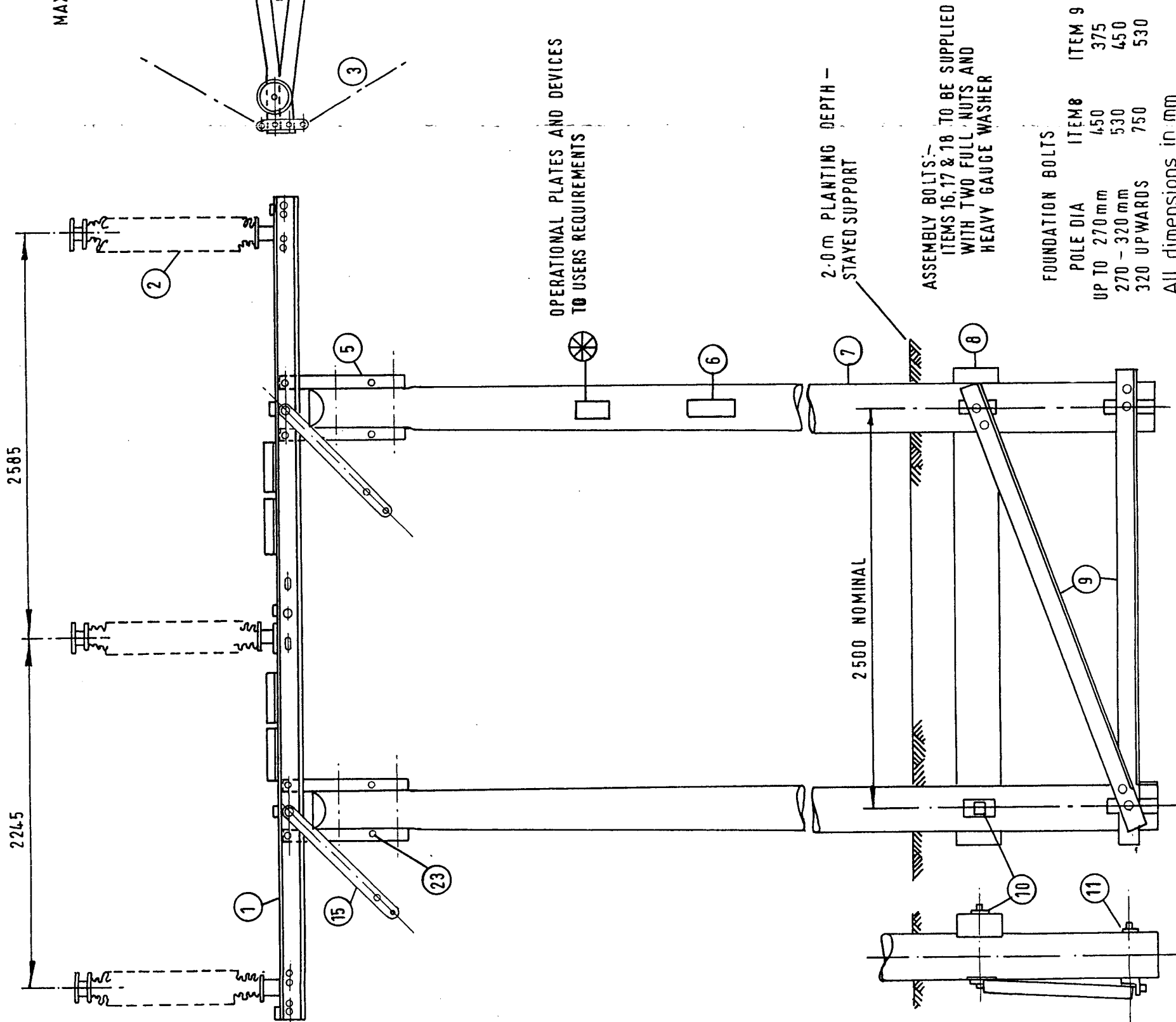
All dimensions in mm

2.0m PLANTING
DEPTH -
STAYED SUPPORT

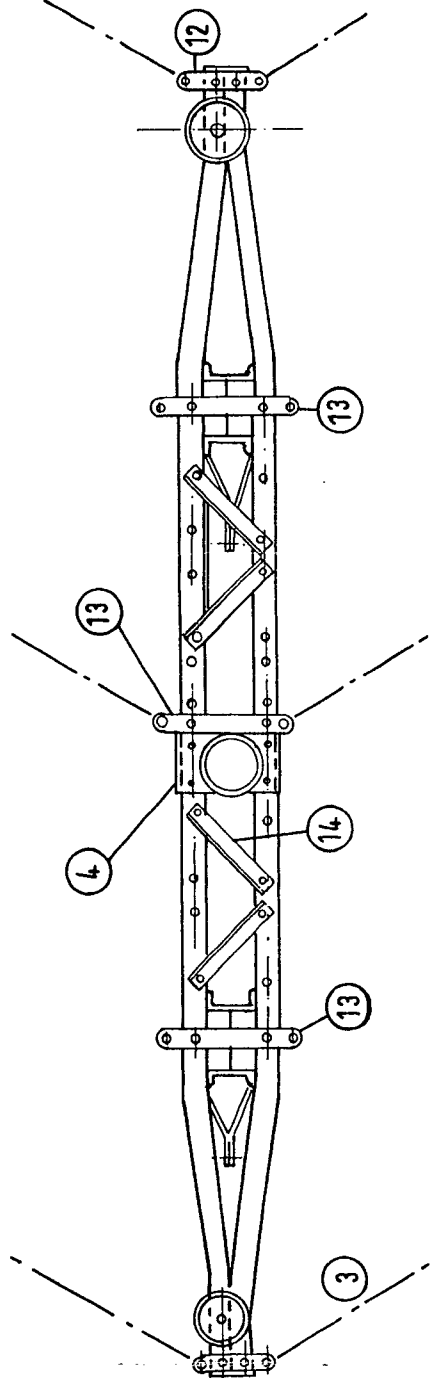
3000 MIN

OPERATIONAL PLATES AND DEVICES
TO USERS REQUIREMENTS

Fig. 9 — General arrangement of angle/section support — single pole



MAXIMUM ANGLE OF DEVIATION 75°



ITEM 13

OUTER LONG TYPE STRAPS ARE FITTED TO BRACE THE CROSSARM AND MUST REMAIN IN POSITION. MAY BE USED FOR STAY ATTACHMENT DURING CONSTRUCTION

No.	ITEM	ESI STD. DWG. No. OR BS. REF.	No. OFF	REMARKS
1	ANGLE CROSSARM	439533	2	
2	POST INSULATOR & CONDUCTOR CLAMP	ESI 43-93	3	
3	TENSION INSULATORS AND	ESI 43-93	6 SETS	
4	INSULATOR FITTINGS	BS 3288	6 SETS	
5	INSULATOR MOUNTING PLATE	439540	1	
6	CROSSARM SUPPORT CHANNEL	439534	4	
7	DANGER PLATE	439701	1	
8	WOOD POLE	435004	2	
9	BRACE BLOCK	439112	1	TYPE 2
10	W-POLE FOUNDATION BRACE	439545	2	
11	SQUARE FLAT WASHER	439604	3	
12	SQUARE CURVED WASHER	439605	2	
13	SHORT TYPE STRAPS	439541	2	
14	LONG TYPE STRAPS	439542	3	
15	CROSS BRACING	439543	4	
16	STAY ASSEMBLY	ESI 43-91		
17	BOLTS M20 X 60	ESI 43-96	39	
18	M24 X 60	"	4	
19	M20 X 375	"	4	
20	M20 X 450	"		AS REQUIRED
21	M20 X 530	"		"
22	M20 X 750	"		"
23	WASHER TAPER - D	"	9	
24	EARTHING ATTACHMENT NUT	439554	1	

Fig. 10 — General arrangement angle/section 'H' pole

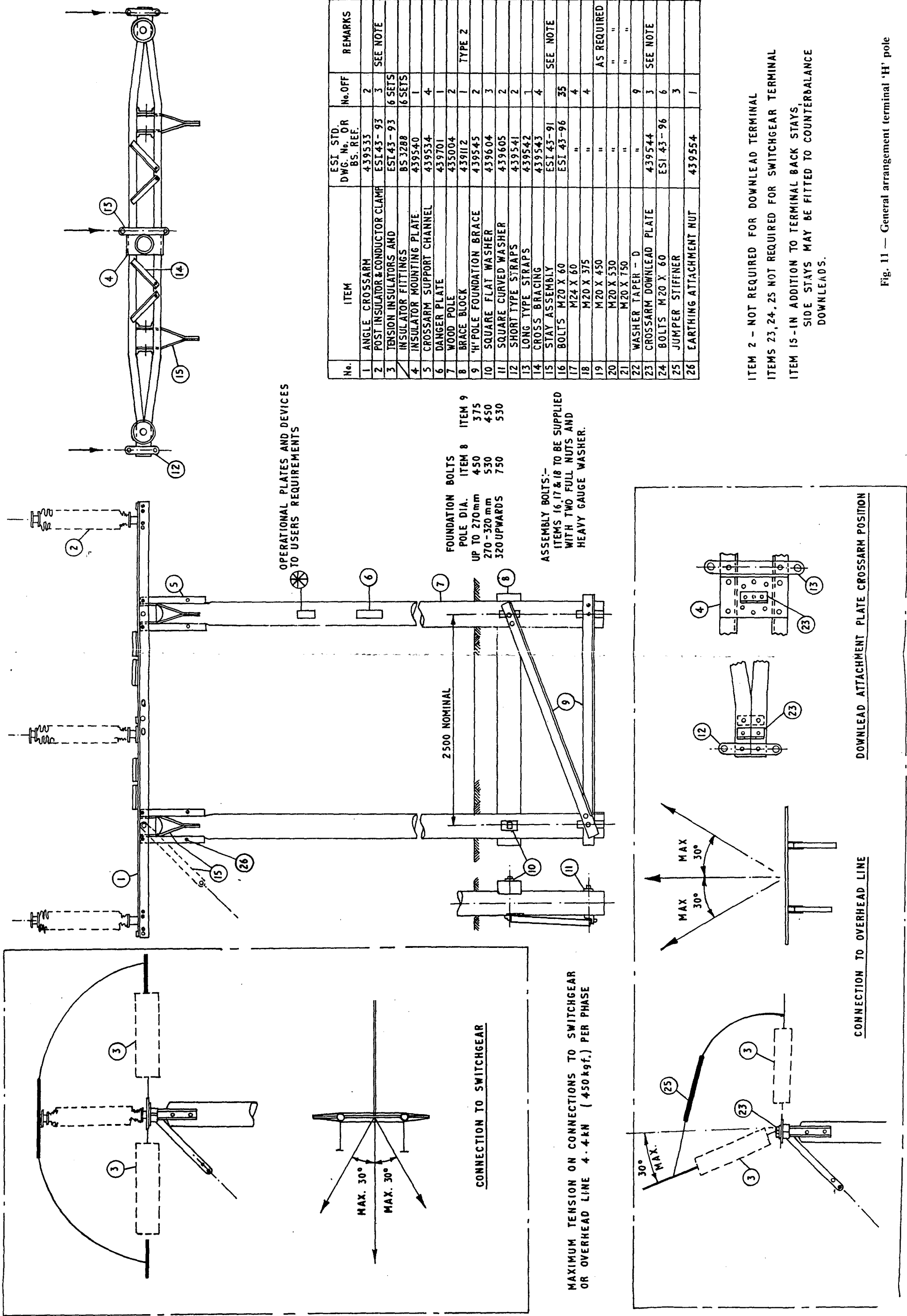
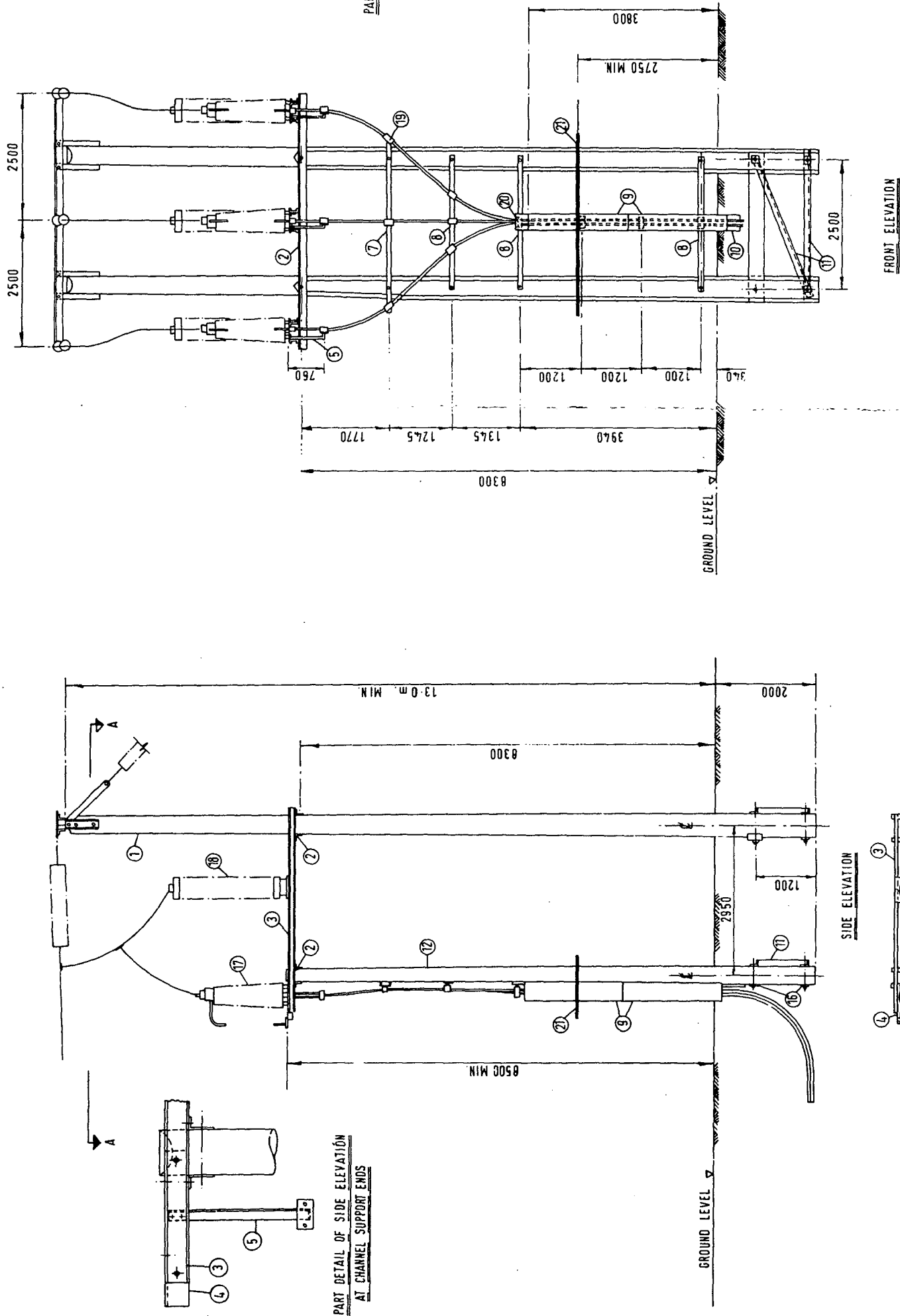


Fig. 11 — General arrangement terminal 'H' pole



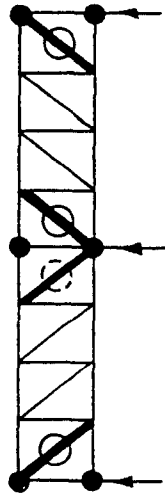
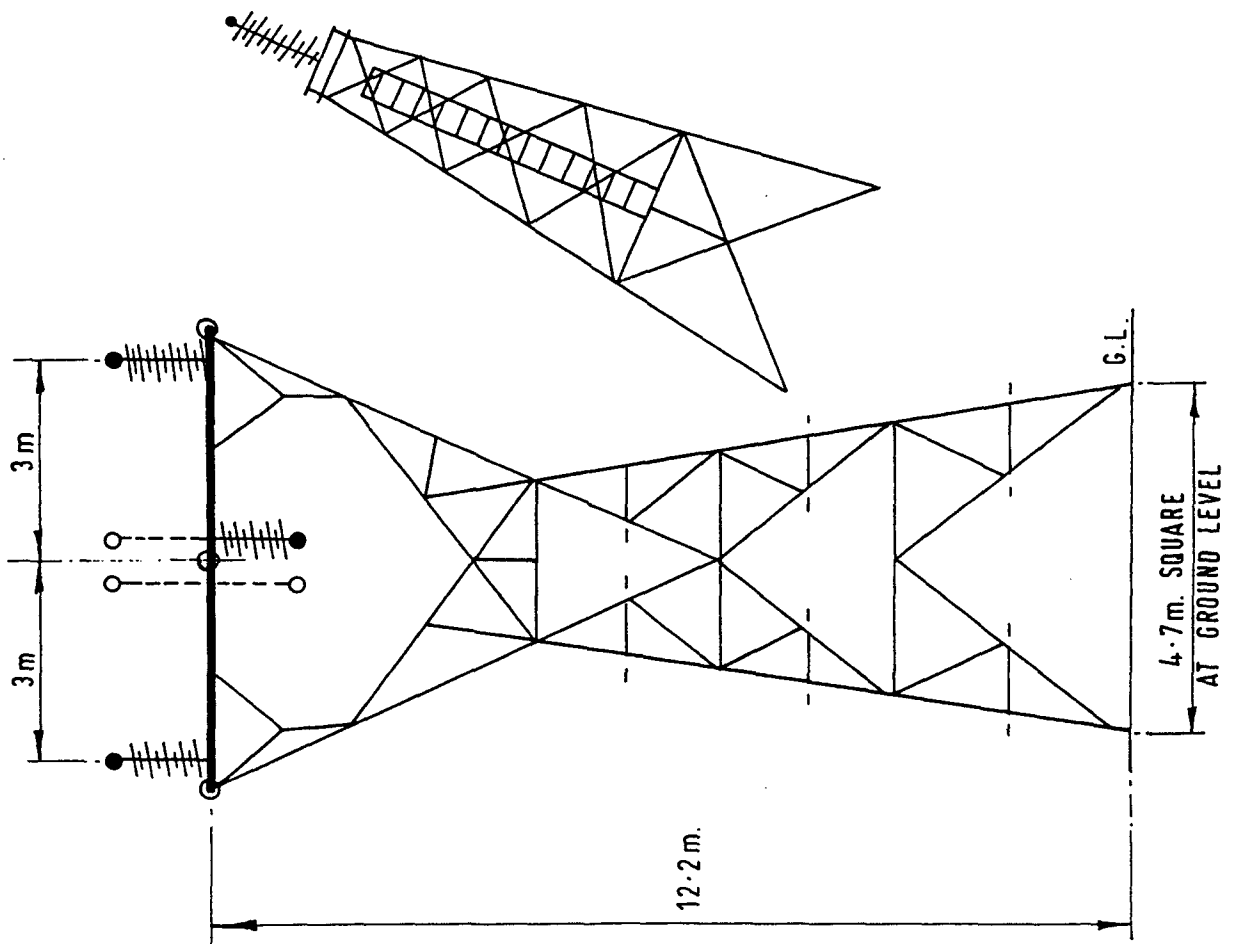
No.	ITEM	ESI STD. ORG. No. OR BSI REF.	No. OFF	REMARKS
1	TERMINAL 'H' POLE	435011	1	SEE NOTES
2	CROSSARM CABLE TERM.	439546	4	
3	CHANNEL SUPPORTS	439547	6	
4	SEALING END SUPPORT	439548	3 SETS	IN HANDED PAIRS
5	CABLE CLEAT SUPPORT	439549	3	
6	SURGE DIVERTER PLATE	439550	3	
7	CABLE SUPPORT CROSSARM 'A'	439551	1	
8	CABLE SUPPORT CROSSARM 'B'	439552	3	
9	CABLE GUARD	439553	2	
10	CABLE BACKBOARD	435014	1	
11	'H' POLE FOUNDATION BRACE	439545	2	
12	CABLE TERMINAL AUX. POLE	435005	2	
13	BOLTS M12 x 60	ESI 43-96	6	
14	M16 x 50	-	54	
15	M20 x 375	-	14	
16	SQUARE CURVED WASHER	439605	3	
17	132kV SEALING END		3	SEE NOTES
18	132kV SURGE DIVERTER		3	
19	SINGLE CABLE CLEAT		9	
20	TREFOIL CABLE CLEAT		4	
21	ANTI-Climbing DEVICE	ESI 43-90	-	

NOTES:-
ITEM 1. TO INCLUDE ALL ITEMS FROM 435011 EXCEPT Nos 2, 23, 24 & 25.

ITEM 17. 132kV SEALING END TO INCLUDE ARCING HORNS AND EARTH END ATTACHMENT PLATES.

All dimensions in mm unless otherwise stated

Fig. 12 — General arrangement pole mounted cable termination



PILOTINSULATORS ARE TO BE POST TYPES
SECURED BY 1 X M20 STUD.

CENTRAL PILOT INSULATOR CAN BE FITTED IN ANY OF
2 INVERTED OR UPRIGHT POSITIONS.

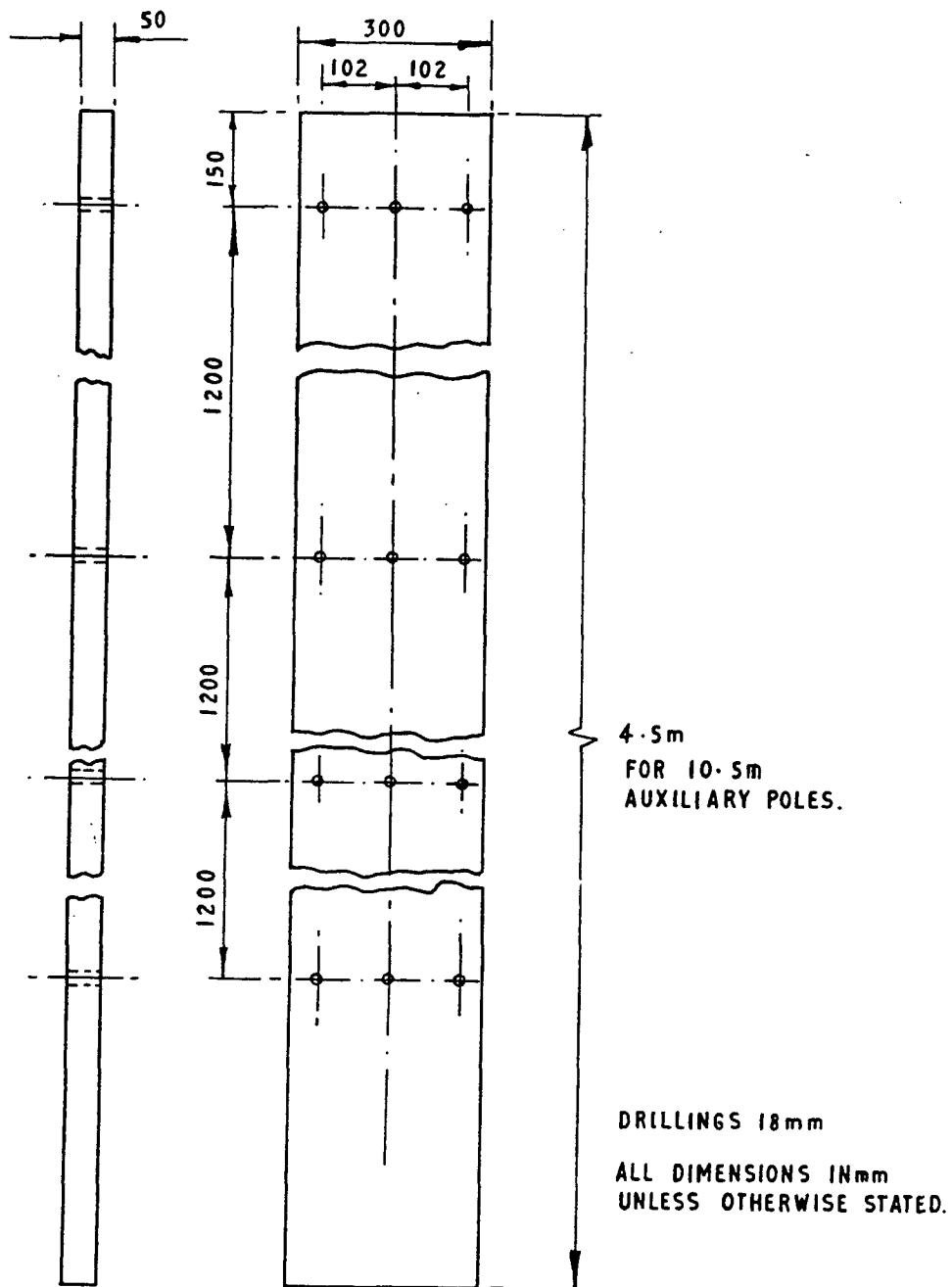
ATTACHMENT POINTS FOR INCOMING LINE AND DOWNLEADS
ARE AS SHOWN ON THE PLAN VIEW.

ADJACENT WOOD POLE SUPPORT TO HAVE 2.5 m.
CONDUCTOR SPACING.

TOWER TO GEG8 DRAWING BK35/25603

Fig. 13 — General arrangement steel tower terminal

ESI Drawing No. 435013



MATERIAL: DEAL WOOD
PRESSURE CREOSOTED

SECTION: 300 x 50

ESI Drawing No. 435014

Fig. 14 — Cable backboard — cable terminal structure

POLE LENGTH (METRES)																						
WINDSPAN REQUIREMENT (METRES)								POLE DIA (mm)	PLANTING DEPTH (METRES)	GROUND NORMAL (CLAY) 200 kN/m ² /m DEPTH				GROUND HARD DRAINED 300 kN/m ² /m DEPTH				GROUND HARD STONEY 400 kN/m ² /m DEPTH				PLANTING DEPTH (METRES)
25	50	75	100	125	150	175	200			A	B	C	D	A	B	C	D	A	B	C	D	
10								230	2.0	2.5	1.5			1.5							2.0	
10.5								235	2.0	2.5	1.5			1.5							2.0	
11								240	2.25	1.5											2.25	
11.5								245	2.25	1.5											2.25	
12								250	2.0	2.5	1.5			1.5							2.0	
13								255	2.25	2.5	1.5			1.5							2.25	
10								260	2.25	2.5	1.5			1.5							2.25	
10.5								265	2.5	2.5	1.5										2.5	
11								270	2.5	2.5	1.5										2.5	
11.5								275	2.75	1.5											2.75	
12								280	3.0												3.0	
13								285														
15								290														
16								295	2.0	2.5	1.5			1.5							2.0	
14								305	2.25	2.5	1.5			1.5							2.25	
11								310	2.25	2.5	1.5			1.5							2.25	
11.5								315	2.5	2.5	1.5										2.5	
12								320	2.5	2.5	1.5			1.5	1.5						2.5	
10								325	2.75	1.5											2.75	
10.5								330	3.0												3.0	
11								335														
13								340														
14								345														
10								350	2.0	2.5	1.5			1.5							2.0	
10.5								355	2.25	2.5	1.5			1.5							2.25	
11								360	2.25	2.5	1.5			1.5							2.25	
11.5								365	2.5	2.5	1.5										2.5	
14								370	2.5	2.5	1.5			1.5							2.5	
10								375	2.75	1.5											2.75	
10.5								380	3.0												3.0	
11								385														
11.5								390														
12								395														
14								400	2.0	2.5	1.5			1.5							2.0	
15								405	2.25	2.5	1.5			1.5							2.25	
13								410	2.25	2.5	1.5			1.5							2.25	
14								415	2.5	2.5	1.5										2.5	
16								420	2.5	2.5	1.5										2.5	
17								425	2.75	1.5											2.75	
15								430	3.0												3.0	
13								435														
16								440														
17								450														
18								465														

FACTOR OF SAFETY = 3.5
FOUNDATION TYPES (NOT TO SCALE)

A

B

C

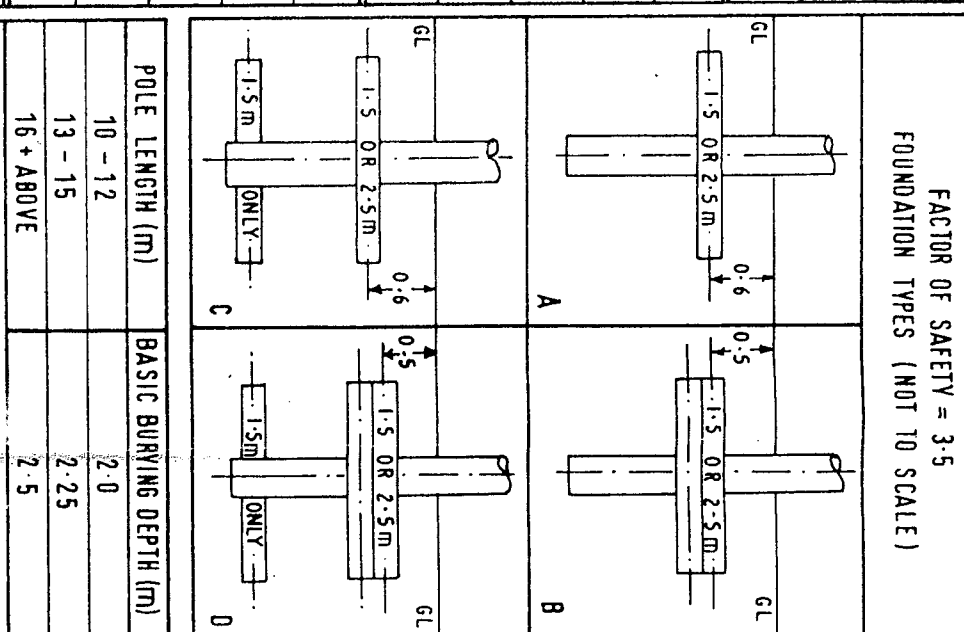
D

POLE LENGTH (m)	BASIC BURYING DEPTH (m)
10 - 12	2.0
13 - 15	2.25
16 + ABOVE	2.5

WINDSPAN CAPACITIES FOR MINIMUM BS 1990 POLE SIZES (METRES)											
Pole Length (m)	10	10.5	11	11.5	12	13	14	15	16	17	18
230	29										
235	29										
240	29										
245	29										
250	29										
260	29										
275	29										
285	64										
290	63										
295	63										
300	62										
305	62										
320	62										
330											
335											
350											
365											
375											
390											

ENTRY OF 1.5 OR 2.5 UNDER A,B,C, OR D, INDICATES LENGTH OF TOP BAULK
ENTRY ☐ INDICATES FOUNDATION CAPACITY OBTAINED AT LESSER DEPTH OR FOUNDATION TYPE.
ENTRY ☒ INDICATES PLANTING DEPTH OR GROUND CLASSIFICATION INSUFFICIENT.
* ANY POLE DIAMETER IN EACH GROUP CAN BE MATCHED WITH ANY PLANTING DEPTH IN THE SAME GROUP.

Table A1 — Windspan/pole/foundation selection chart — single supports for lines in severe environment



WINDSPAN CAPACITIES FOR MINIMUM BS 1990 POLE SIZES (METRES)											
POLE LENGTH (m)	BASIC BURYING DEPTH (m)										
10-12	2.0										
13-15	2.25										
16+ ABOVE	2.5										
10	10.5	11	11.5	12	13	14	15	16	17	18	
230	29	29									
235	29	29									
240	29	29									
245	29	29									
250	29	29									
260	29	29									
275	32										
285	35										
290	35										
295	35										
300	39										
305	39										
320	42										
330	42										
335	42										
350	70										
365	73										
375	72										
390	75										